

Piezoelectric Element and Contact Stress (20 points)

Introduction

Some non-conducting crystalline materials, such as quartz and lead zirconate titanate, show an electrical response when mechanical pressure is applied. This phenomenon is called the piezoelectric effect. Mechanical stress polarizes the crystal and produces charge separation. One may understand this by considering that deformation gives each molecule an electric dipole moment. Conversely, an applied electric field can also produce mechanical stress; this is called the inverse piezoelectric effect. In this problem, the inverse piezoelectric effect may be neglected.

In this problem we study a simple device using the piezoelectric effect: a piezoelectric element. The piezoelectric material is sandwiched between 2 circular metal plates. When a force perpendicular to the plates is applied, a voltage corresponding to that force appears between the two plates.

Apparatus List

- A: Piezoelectric element with lead wires. It is mounted between a wooden base plate and a top plate. The top plate has a small hole exposing part of the upper electrode of the piezoelectric element. This electrode is thin and easily bent.
- B: Digital multimeter. Its internal resistance is given in the table below.
- C: A 1.5 V AA battery and its connector.
- D: Capacitor of unknown capacitance. A diode is soldered to one terminal. When a forward current flows, the voltage drop across the diode is 0.56 V.
- E: 4 push-button switches that conduct only while pressed.
- F: 6 crocodile clips.
- G: Digital scale, maximum load 10 kg.
- H: Digital stopwatch.
- I: 2 rubber balls.
- J: Wooden stand with an adjustable-height release mechanism.
- K: 50 cm ruler.
- L: 2 short wooden rods of diameter 2 mm.
- M: Large metal screw.
- N: Wooden clothespin.
- O: Pencil, pen, and pencil sharpener.

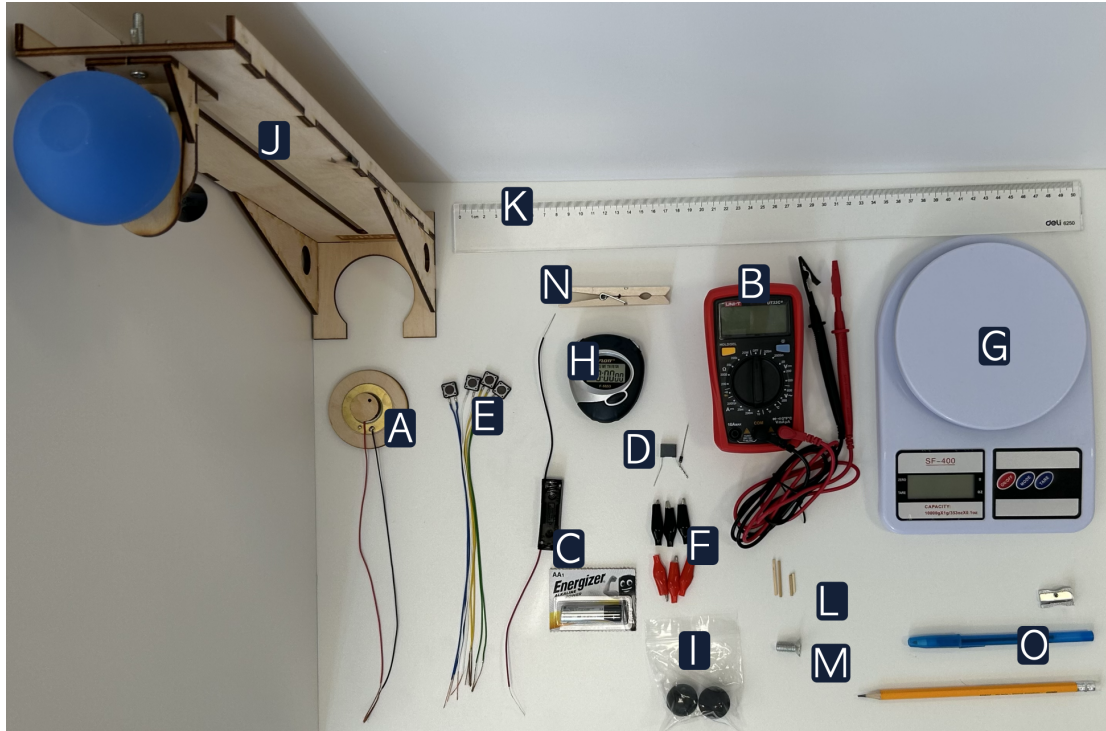


Figure 1: Overall view of the apparatus used in the original experiment.

About the Apparatus

Figure 1 shows the complete set of apparatus used in the original experiment. Although no actual measurements are performed in this drill, understanding the role of each item helps when designing circuits and reasoning about the measurement principles.

- **Piezoelectric element A**

This element consists of piezoelectric material between 2 metal electrodes. When a vertical force is applied, charge appears on the electrodes and a voltage is produced. In this problem, treat the piezoelectric element as an element with capacitance C_p that produces charge when a force is applied.

- **Wooden top plate**

A wooden top plate is attached above the piezoelectric element. It transmits the applied force to the whole piezoelectric element.

- **Digital multimeter B**

It is used to measure voltages in the circuit. In DC voltage mode it has a finite internal resistance R_V . Its value is shown in the table below.

Range	Internal resistance R_V
2 V	11.1 M Ω
200 mV	9.65 M Ω

- **Battery C**

It is used to charge a capacitor or the piezoelectric element to a known voltage U_0 . In the measurement of the capacitance C_p of the piezoelectric element, it provides approximately the same initial condition in each trial.

- **Capacitor D**

The charge and voltage generated by the piezoelectric element are transient and difficult to read directly. Therefore, charge is transferred from the piezoelectric element to a known or unknown capacitor and read as a more slowly varying voltage.

- **Capacitor with diode**

A diode allows current to flow only in one direction. It can therefore prevent charge that has flowed from the piezoelectric element into the capacitor from flowing back.

- **Push-button switches E**

They are used to switch circuit connections. They are needed, for example, to carry out operations such as charging the piezoelectric element, disconnecting it from the battery, transferring charge to a capacitor, and discharging before the next measurement.

- **Stand with release mechanism J**

This mechanism releases a rubber ball from a known height with as little extra initial velocity as possible. By changing the height, the mechanical energy mgh just before collision can be controlled. In this problem, consider using this mechanism to study the elasticity of the ball and the maximum force during collision.

- **Ruler K**

This provides the length scale for reading the drop height and rebound height. In the part on ball elasticity it is used to compare h and the rebound height h_b .

- **Scale G**

It is used to measure mass and convert it to force $F = mg$. In calibrating the force-voltage response of the piezoelectric element, the force applied by a known mass is used as a reference.

Notes on Analysis

- Unless explicitly requested, uncertainty analysis is not required.
- When using a fit or an approximation, choose an appropriate data range if necessary. In that case, state clearly which range was used.

Part A : Measurement of the coefficient of restitution (2.3 points)

Consider dropping the more elastic rubber ball onto a rigid horizontal surface, not onto the piezoelectric element. The ball is released gently from height h above the floor, and the height of the highest point of the first rebound is denoted by h_b . The coefficient of restitution e is defined from the speed just before and just after the collision. Offsets such as the radius of the ball may be ignored; in other words, h and h_b may be regarded as the vertical displacement of the center of mass of the ball.

For each height, the rebound height h_b was measured 5 times. The results are shown in **Table A-1**. Let $\overline{h_b}$ be the mean value for each height, and define the uncertainty of this mean, δh , by

$$\delta h = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^N (h_{b,i} - \overline{h_b})^2}$$

where $N = 5$.

Data A-1: Drop height and rebound height

h [mm]	h_b [mm]				
120	82	94	104	89	101
180	129	151	138	148	143
240	178	195	208	186	199
300	223	241	257	232	247
360	270	292	304	280	297

A.1 Express the coefficient of restitution e in terms of h and h_b . 0.2pt

A.2 For each drop height, calculate the mean rebound height $\overline{h_b}$ and its uncertainty δh . 1.0pt

A.3 Determine the energy retention ratio η of the ball before and after the collision by plotting the results of **A.2**. Include error bars and give η with its uncertainty. Assume no uncertainty in h . 0.8pt

A.4 Using the result of **A.3**, evaluate the coefficient of restitution e with uncertainty. 0.3pt

Part B : Measurement of the capacitance C (2.9 points)

We wish to determine the capacitance C of capacitor D, whose capacitance is unknown. No external discharge resistor is provided. However, when the digital multimeter B is used as a DC voltmeter, the capacitor discharges through the internal resistance R_V of the meter. Design an experiment that uses this property to measure the discharge curve of the capacitor.

First charge the capacitor using battery C. Then remove the battery, connect the multimeter as a DC voltmeter, and measure the voltage $U(t)$ across the capacitor. The multimeter is kept on the 2V range throughout the measurement; use the value of R_V given in the table at the beginning. The battery voltage was measured to be

$$U_0 = U(0) = 1.405 \text{ V}$$

B.1 Draw a circuit diagram that can perform the charging and discharging process needed to measure the unknown capacitor C , and explain the procedure. If more than one switch is used, label them $S_1, S_2, \dots$ and match the labels to the procedure. 0.7pt

B.2 Express the voltage $U(t)$ read by the multimeter using U_0 and any necessary parameters. Show the derivation. 0.6pt

Table B-1 shows voltage data obtained by the method above.

Data B-1: Measured discharge curve

t [s]	0	5	10	15	20	25	30	35	40
$U(t)$ [mV]	1405	1151	943	773	634	518	426	342	283

t [s]	45	50	55	60	65	70	75	80	85
$U(t)$ [mV]	235	191	158	124	103	83	69	57	46

B.3 Linearize the data in **Table B-1** appropriately, draw a graph, and determine the capacitance C . Uncertainty analysis is not required. 1.6pt

Part C : Measurement of the capacitance C_p of the piezoelectric element (2.9 points)

The piezoelectric element not only produces charge when a force is applied, but also behaves as a dielectric sandwiched between 2 electrodes and hence has a capacitance C_p . Since C_p is about 2 orders of magnitude smaller than C , it is difficult to measure accurately by the direct discharge method of Part B. Charge the piezoelectric element to a known voltage U_0 , and repeatedly transfer part of its charge to the known capacitor C . One complete measurement for obtaining V_n has the following structure:

discharge C at the start of the measurement
 $\rightarrow$ repeat the same charge-transfer loop n times
 $\rightarrow$ measure V_n

The measurement is performed in this order. Here, one loop consists of charging the piezoelectric element, connecting it to the known capacitor to transfer charge, and discharging the piezoelectric element. The known capacitor C is not discharged during the n loops, but accumulates charge. First discharge C sufficiently so that $V_0 = 0$; after repeating the same loop n times, measure the voltage V_n across the known capacitor only once. For example, $n = 3$ means that the charge-transfer loop is repeated 3 times before reading V_3 .

The circuit diagram is shown in **Figure 2**. S_1 is used to charge the piezoelectric element, S_2 to transfer charge from the piezoelectric element to the known capacitor, S_3 to measure the voltage across the known capacitor, and S_4 to discharge the piezoelectric element. To discharge the known capacitor C at the beginning, keep S_3 closed long enough so that it discharges through the internal resistance of the voltmeter.

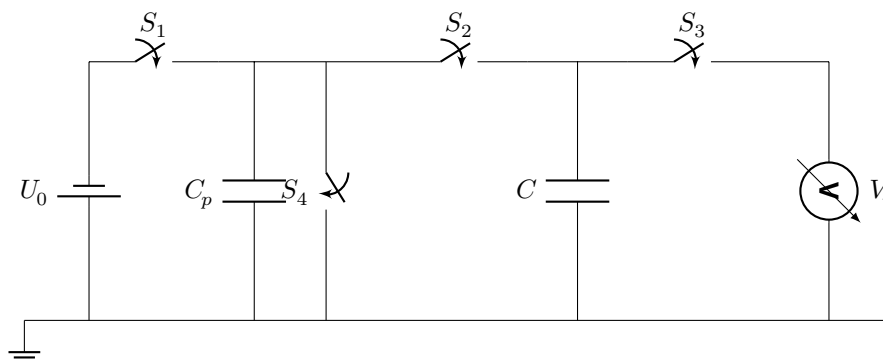


Figure 2: Circuit for measuring the capacitance C_p of the piezoelectric element.

- C.1** Using **Figure 2**, write the complete measurement sequence for obtaining V_n on the answer sheet. Distinguish the initial discharge, the charge-transfer loop repeated n times, and the final voltage measurement. Write each step in the form "set/keep S_i " 0.6pt

open/closed”, and make clear that the known capacitor C is not discharged during the loop.

C.2 Let V_n be the voltage appearing across the known capacitor C after n loops. Express V_n in terms of the battery voltage U_0 , C , C_p , and n . Show the derivation. Assume that $V_0 = 0$. 0.6pt

Table C-1 shows data obtained by this method.

Data C-1: Number of loops and voltage V_n

n	1	2	3	4	5	6	8	10	12	15	20	25	30
V_n [mV]	13.3	26.5	43.0	53.9	69.8	82.5	107	135	159	198	260	314	371

C.3 Using **Table C-1**, draw an appropriate graph and determine the capacitance C_p of the piezoelectric element. Use the values of U_0 and C obtained in Part B. You may use $C \gg C_p$ if needed. 1.7pt

Part D : Force-voltage response of the piezoelectric element (5.0 points)

In this part, investigate the relation between the force F applied to the piezoelectric element and the voltage V generated by the piezoelectric element itself. It is difficult to connect a voltmeter directly to the piezoelectric element and read this voltage. Instead, transfer the charge generated by the piezoelectric element to the known capacitor C and read the voltage U across that capacitor. In this part, take $g = 9.81 \text{ m s}^{-2}$.

- D.1** Give 2 reasons why directly connecting the piezoelectric element to a voltmeter and reading the voltage does not work well. One of the reasons must use the term time constant. 0.4pt

Figure 3 shows the circuit used to measure the force-voltage response. Let V be the voltage generated by the piezoelectric element, and let U be the voltage held on the known capacitor C and read by the voltmeter. The circuit uses 4 switches. In particular, S_2 and S_4 are switches for discharging the piezoelectric element and the known capacitor, respectively.

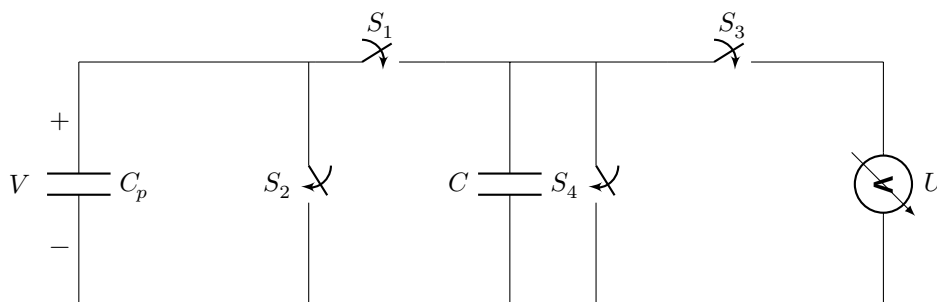


Figure 3: Circuit for measuring the force-voltage response of the piezoelectric element.

- D.2** Draw a simple diagram showing the arrangement of the piezoelectric element, the scale, and the way the force is applied. Then list the measurement procedure using the circuit in **Figure 3**. The procedure must make it possible to read the value of V corresponding to a particular instantaneous value of the force F applied to the piezoelectric element, and must include the operation for discharging the piezoelectric element and the capacitor before measuring another value. Briefly explain why U must be read quickly after opening/closing the switch for voltage readout. 1.1pt

- D.3** In the circuit of **Figure 3**, give the formula relating the readout voltage U to the voltage V of the piezoelectric element. Also give the formula relating the scale reading M to the force F . 0.3pt

Table D-1 shows the measurement results. Here M is the scale reading, and U is the voltage read across the capacitor in **Figure 3**.

Data D-1: Scale reading M and readout voltage U

M [g]	U [mV]	M [kg]	U [mV]	M [kg]	U [mV]
10	7.6	0.86	308	2.65	862
30	8.6	0.93	322	2.82	882
70	27.8	1.01	357	3.05	868
110	44.9	1.09	390	3.28	889
140	45.3	1.18	412	3.55	856
180	65.6	1.27	450	3.82	878
220	82.7	1.34	479	4.10	893
250	83.1	1.43	502	4.48	865
290	103.4	1.55	541	4.86	881
330	122.6	1.64	590	5.22	860
360	124.0	1.73	612	5.60	886
400	145.2	1.85	672	5.98	869
450	157.8	1.96	696	6.31	883
490	178.0	2.08	745	6.68	857
540	186.5	2.17	752	7.02	885
580	207	2.26	786	7.41	866
630	228	2.34	793	7.83	890
690	238	2.41	824	8.22	864
740	264	2.46	819	8.58	879
800	280	2.50	836	9.00	892

D.4 Convert the data in **Table D-1** to F and V , and plot them on a graph covering a sufficiently wide range. You do not need to convert and plot all data points. Choose points so that the overall trend is clear. 1.8pt

D.5 For small forces, $V = \beta F$ holds. Choose the linear region appropriately, plot it again on a separate graph, and determine the proportionality constant β . 0.8pt

The effective area of the piezoelectric element is $A = 1.33 \times 10^{-4} \text{ m}^2$.

D.6 Using the results of **D.4** and **D.5**, together with C_p obtained in Part C, calculate the saturation voltage V_{sat} , the corresponding force F_{sat} , the saturation pressure 0.6pt

$$p_{\text{sat}} = \frac{F_{\text{sat}}}{A}$$

and the maximum surface charge density

$$\sigma_{\text{max}} = \frac{C_p V_{\text{sat}}}{A}$$

Part E : Ball deformation and contact stress (4.5 points)

Drop the more elastic rubber ball onto the piezoelectric element, and suppose that the ball deforms elastically during the collision. Assume that the relation between the normal force F on the ball and the deformation x is

$$F = kx^\alpha$$

where k and α are constants. Also assume that the linear calibration obtained in Part D, $V = \beta F$ holds. The mass of the ball is $m = 8.0 \text{ g}$ and the gravitational acceleration is $g = 9.81 \text{ m s}^{-2}$.

- E.1** Let $x_{\max}$ be the deformation when the ball is maximally compressed during the collision. Express the maximum voltage $V_{\max}$ indicated by the piezoelectric element in terms of $x_{\max}$. 0.3pt

- E.2** Consider the process in which the ball rebounds after deforming up to $x_{\max}$. By considering the work

$$\int F dx$$

done by the normal force on the ball, express the elastic energy E_{el} stored up to the instant of rebound in terms of k , $x_{\max}$, and α . 0.3pt

- E.3** Assume that the elastic energy E_{el} equals the initial mechanical energy multiplied by the energy retention ratio η obtained in Part A,

$$E_{\text{el}} = \eta mgh$$

. If the relation between the maximum voltage $V_{\max}$ and the drop height h is

$$V_{\max} = ch^\gamma$$

, express the exponent γ in terms of α . 0.5pt

- E.4** Express the relation between the proportionality constant c and k using the necessary parameters. 0.5pt

The collision occurs in a very short time, so the maximum voltage cannot be read directly by the method of Part D. Therefore, based on the circuit in **Figure 3**, add a diode so that charge flows only from the piezoelectric element toward the known capacitor C , thereby holding the maximum voltage.

E.5 To measure $V_{\max}$ during the collision process, where and in which direction should a diode be inserted in the circuit of **Figure 3**? Draw the complete circuit diagram. 0.5pt

Table E-1 shows the readout voltage U obtained when the ball was dropped from various heights h . Five measurements were made at each height.

Data E-1: Drop height h and readout voltage U

h [mm]	U [mV]				
15	79	81	73	73	78
30	120	113	104	107	112
45	144	138	136	137	139
60	171	177	180	192	177
80	204	191	204	199	194
100	222	230	220	230	234
130	269	269	244	258	249
160	292	284	285	288	278
190	304	296	325	298	303
220	303	307	316	340	340
250	356	342	380	336	338
280	384	360	380	392	375
310	397	390	378	335	369
340	411	382	401	415	436
360	397	413	427	432	397

E.6 Determine the value of α from the data in **Table E-1**. Use η from Part A, β from Part D, and the results of Parts B and C if needed. 1.6pt

E.7 Determine the value of k . The unit may be written as $\text{N m}^{-\alpha}$. You may draw another graph if needed. Use the necessary constants from the previous parts. 0.8pt

Is this value close to $\alpha = \frac{3}{2}$, which is expected from Hertzian contact stress? If not, freely discuss why such a difference may appear.

Part F : Contact time (2.4 points)

In this part, assume that the time dependence of the normal force $F(t)$ during collision remains geometrically similar when the impact speed changes. Define a representative contact time by

$$\tau_{\text{eff}} = \frac{\Delta p}{F_{\text{max}}}$$

where Δp is the magnitude of the momentum change due to the collision, and F_{max} is the maximum normal force during the collision.

F.1 A ball with coefficient of restitution e is dropped from height h , and the maximum readout voltage is U_{max} . Express τ_{eff} using known parameters and U_{max} . Use the results of Parts C and D if needed. 0.6pt

F.2 Using the value of e obtained in Part A, calculate τ_{eff} for each height from the data in **Table E-1**. Use the average of the five measured values at each height. Also determine the speed just before collision, $v = \sqrt{2gh}$. 0.7pt

Next, use the other ball. This ball is assumed to rebound negligibly after collision, so take its coefficient of restitution to be $e = 0$. The mass of this ball is also $m = 8.0 \text{ g}$. **Table F-1** shows the maximum readout voltage measured for this ball in the same way.

Data F-1: Maximum readout voltage for the other ball

h [mm]	U [mV]				
15	74	72	72	73	72
30	113	122	107	120	107
70	168	176	159	174	199
110	224	218	257	230	231
130	279	271	253	272	282
170	319	315	316	320	288
190	325	340	339	340	314
210	337	346	360	348	349
250	388	388	397	384	385
290	424	421	355	398	385
330	462	440	451	456	455

F.3 For **Table F-1**, do the same calculation as in **F.2**: determine τ_{eff} and the speed v just before collision for each height. 0.5pt

F.4 For the two balls, plot the relation between τ_{eff} and v on the same graph. State whether τ_{eff} becomes longer, shorter, or remains almost unchanged as v increases. 0.6pt

Piezoelectric Element and Contact Stress (20 points)

Part A : Measurement of the coefficient of restitution (2.3 points)

A.1 (0.2 pt)

$e =$

A.2 (1.0 pt)

$h =$

$h =$

$h =$

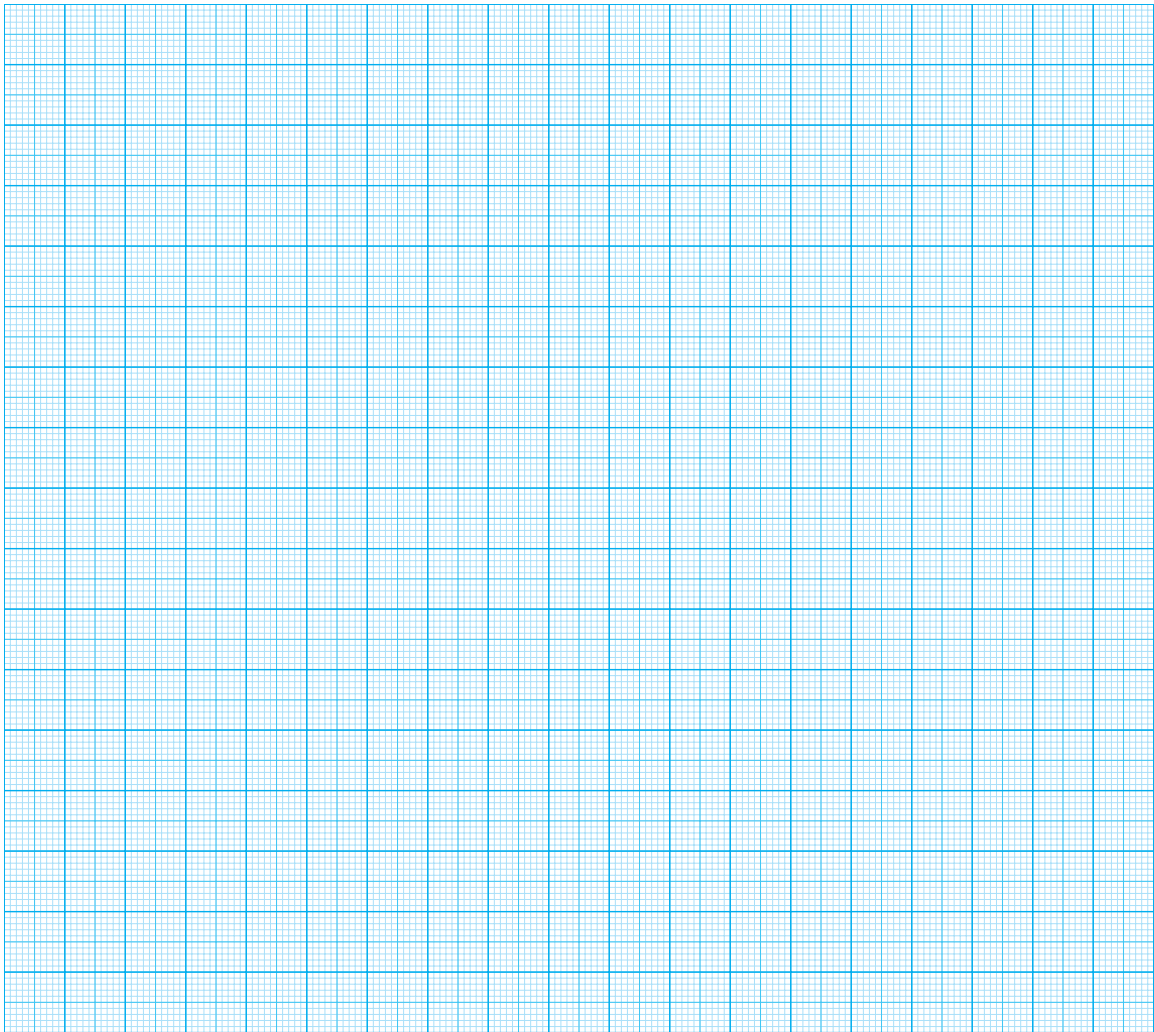
$h =$

$h =$

h [mm]	120	180	240	300	360
$\bar{h}_b$ [mm]					
δh [mm]					



A.3 (0.8 pt)



$\eta =$ $\pm$

A.4 (0.3 pt)

$e =$ $\pm$

Part B : Measurement of the capacitance C (2.9 points)

B.1 (0.7 pt)

Circuit diagram:

Procedure:

No.	Operation
0	Example: close S_1 to discharge capacitor C
1	

B.2 (0.6 pt)

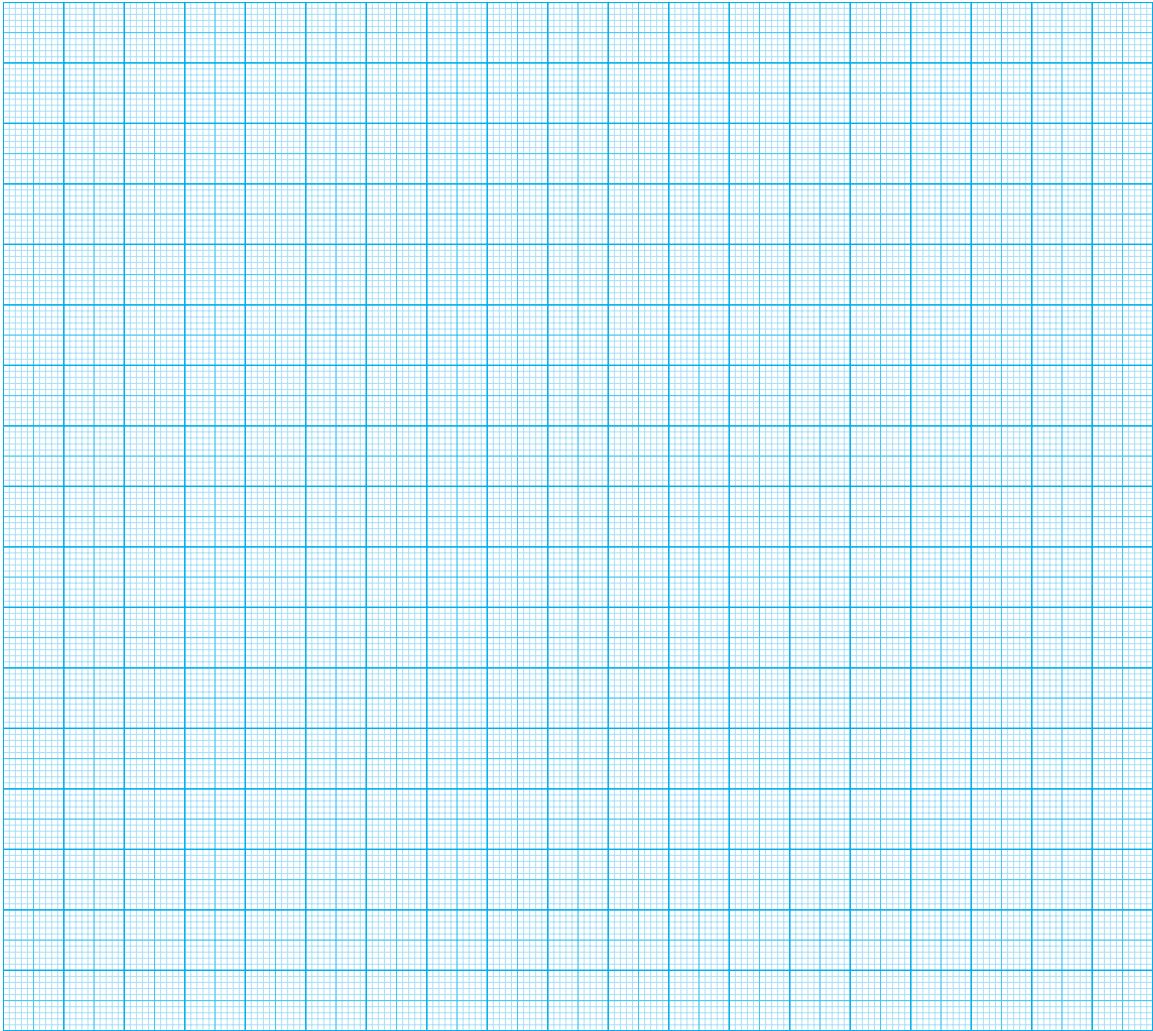
$U(t) =$

B.3 (1.6 pt)

B.3 (cont.) (0 pt)

t [s]	0	5	10	15	20	25	30	35	40
$U(t)$ [mV]	1405	1151	943	773	634	518	426	342	283

t [s]	45	50	55	60	65	70	75	80	85
$U(t)$ [mV]	235	191	158	124	103	83	69	57	46



$C =$

Part C : Measurement of the capacitance C_p of the piezoelectric element (2.9 points)

C.1 (0.6 pt)	
Stage	Operation
$V_0 = 0$	
Loop	
Measurement	

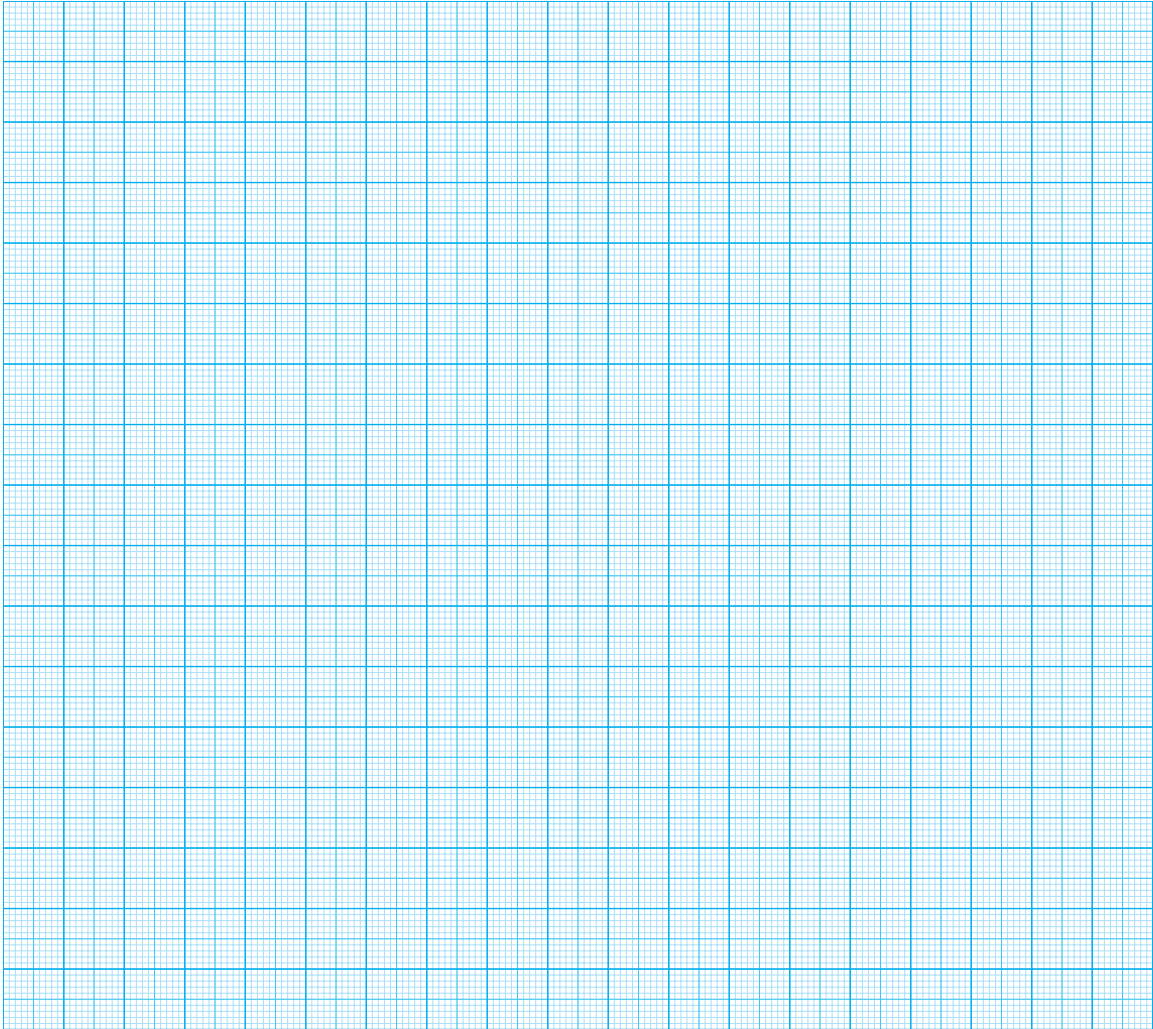
C.2 (0.6 pt)
$V_n =$

C.3 (1.7 pt)
If a transformation or approximation is needed, show it here:

C.3 (cont.) (0 pt)

n	1	2	3	4	5	6	8
V_n [mV]	13.3	26.5	43.0	53.9	69.8	82.5	107

n	10	12	15	20	25	30
V_n [mV]	135	159	198	260	314	371



$C_p =$

Part D : Force-voltage response of the piezoelectric element (5.0 points)

D.1 (0.4 pt)

Reason 1:

Reason 2:

D.2 (1.1 pt)

Arrangement and application of force:

Measurement procedure:

No.	Operation
1	

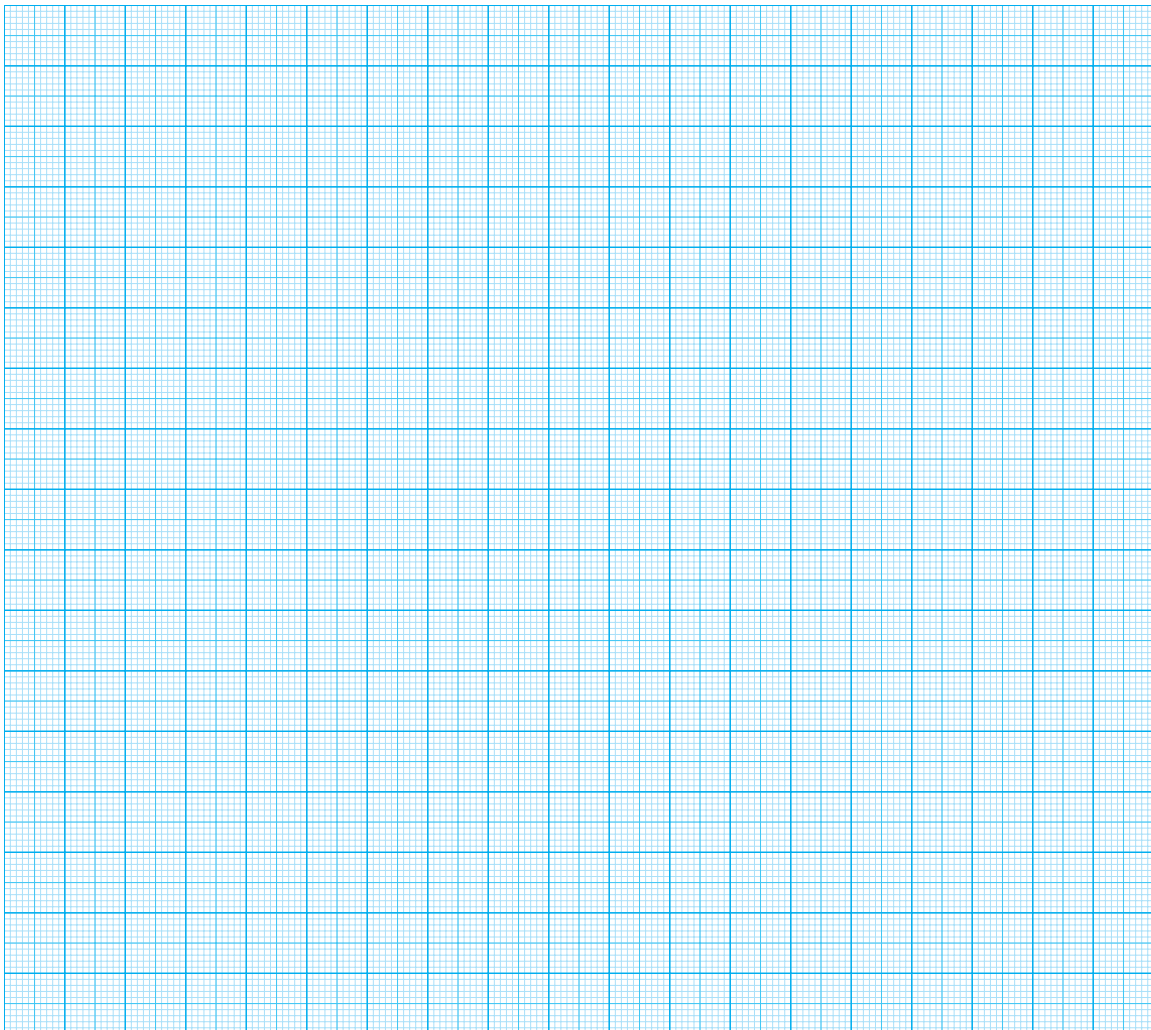
Reason:

D.3 (0.3 pt)

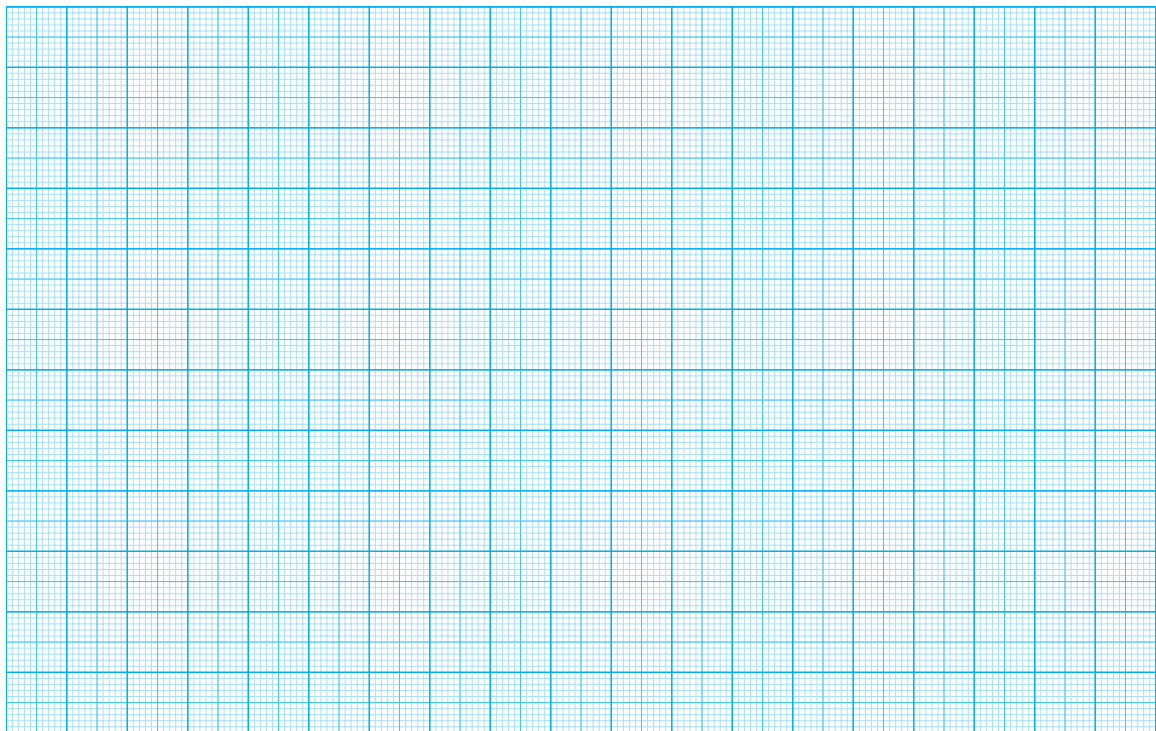
$V =$

$F =$

D.4 (1.8 pt)

[illegible][illegible]

D.5 (0.8 pt)



$\beta =$

D.6 (0.6 pt)

$V_{\text{sat}} =$, $F_{\text{sat}} =$
 $p_{\text{sat}} =$, $\sigma_{\text{max}} =$

Part E : Ball deformation and contact stress (4.5 points)

E.1 (0.3 pt)

$V_{\max} =$

E.2 (0.3 pt)

$E_{\text{el}} =$

E.3 (0.5 pt)

$\gamma =$

E.4 (0.5 pt)

$c =$

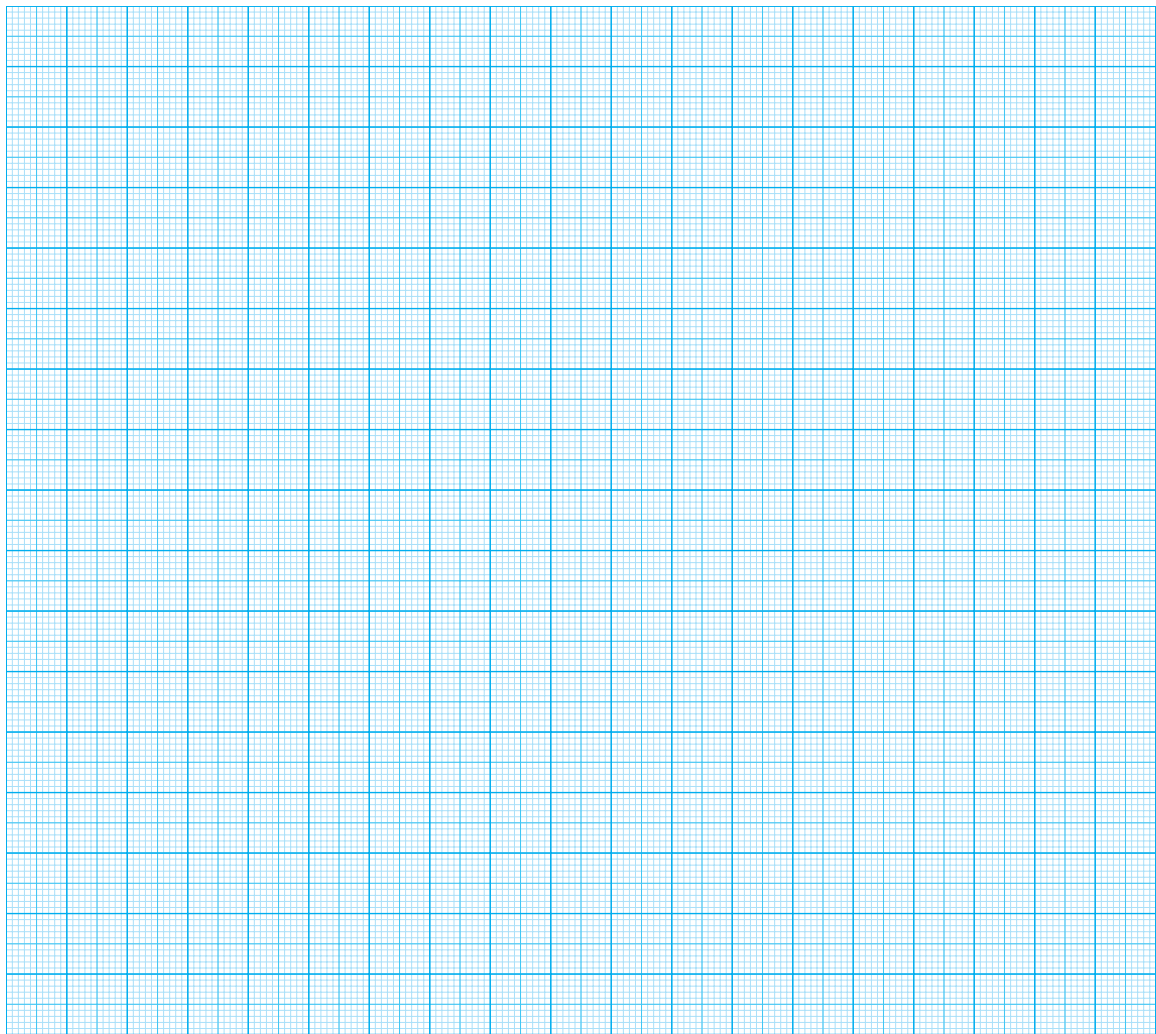
E.5 (0.5 pt)

Circuit diagram:

E.6 (1.6 pt)

h [mm]	U [mV]					U_{max} [mV]	V_{max} [V]			
15	79	81	73	73	78					
30	120	113	104	107	112					
45	144	138	136	137	139					
60	171	177	180	192	177					
80	204	191	204	199	194					
100	222	230	220	230	234					
130	269	269	244	258	249					
160	292	284	285	288	278					
190	304	296	325	298	303					
220	303	307	316	340	340					
250	356	342	380	336	338					
280	384	360	380	392	375					
310	397	390	378	335	369					
340	411	382	401	415	436					
360	397	413	427	432	397					

E.6 (cont.) (0 pt)



$\alpha =$

E.7 (0.8 pt)

$k =$

Part F : Contact time (2.4 points)

F.1 (0.6 pt)

$\tau_{\text{eff}} =$

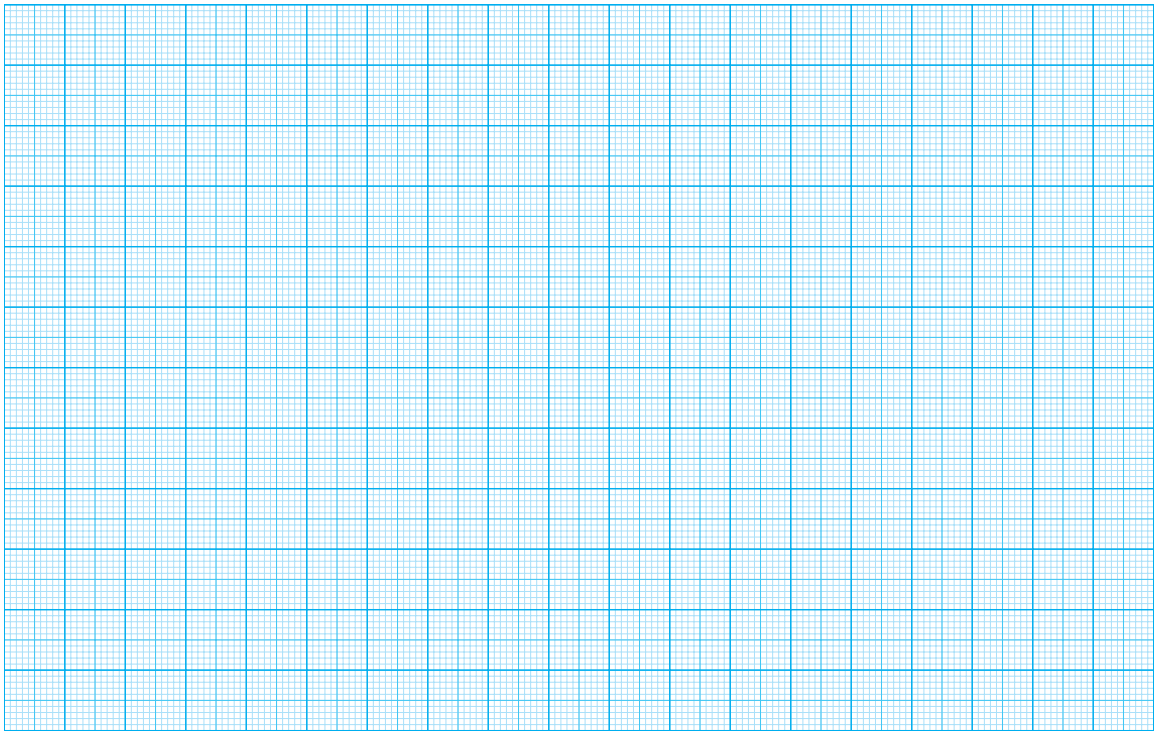
F.2 (0.7 pt)

h [mm]	v [m/s]	U_{max} [mV]	τ_{eff} [ms]		
15					
30					
45					
60					
80					
100					
130					
160					
190					
220					
250					
280					
310					
340					
360					

F.3 (0.5 pt)

h [mm]	v [m/s]	U_{max} [mV]	τ_{eff} [ms]		
15					
30					
70					
110					
130					
170					
190					
210					
250					
290					
330					

F.4 (0.6 pt)

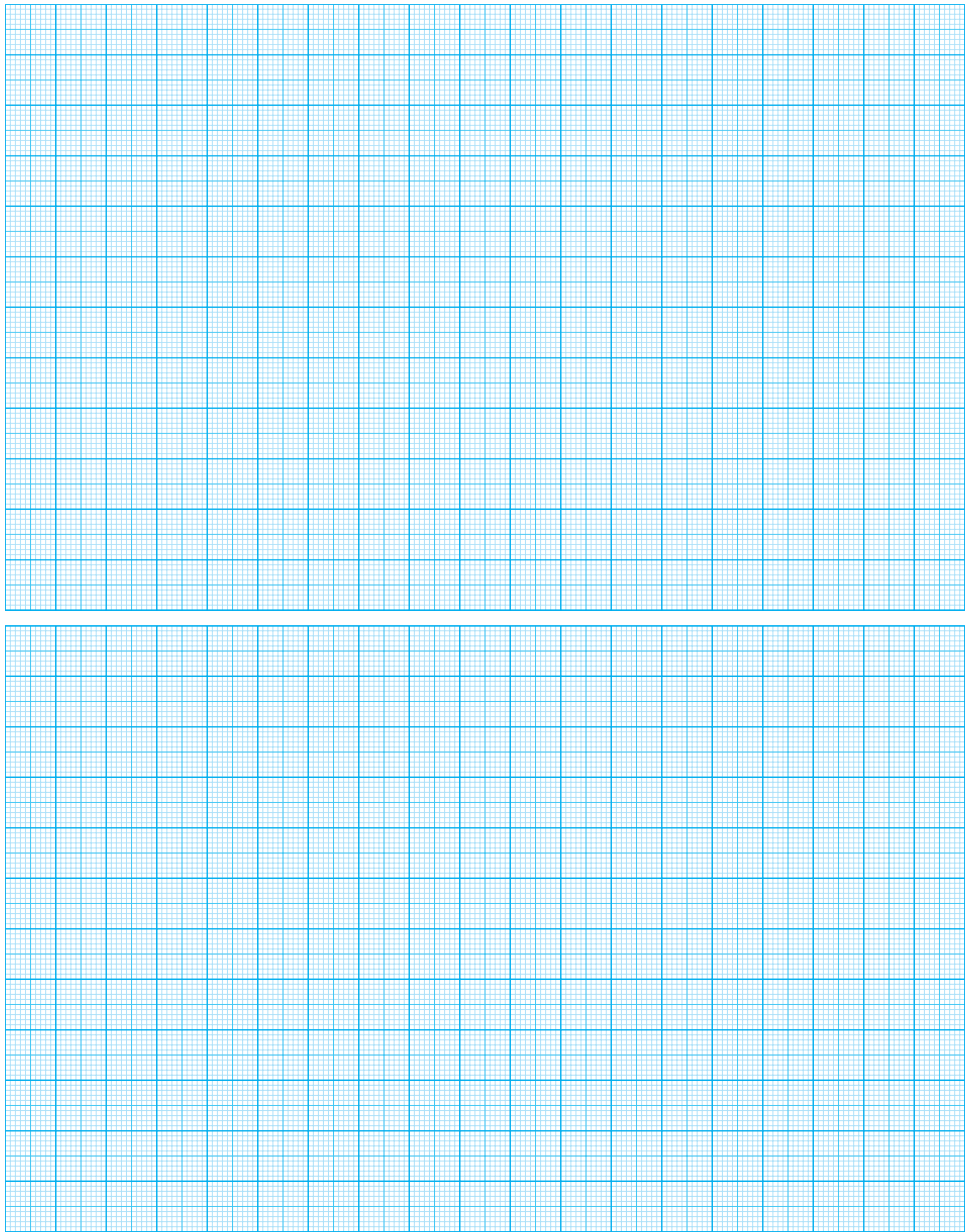


Conclusion:



Extra graph paper

Extra graph paper (0 pt)



Common Marking Scheme

Marking Scheme

Points for graphs:	0.4
Axis labels or units are missing (arrows are not required)	-0.1
Appropriate ticks are missing	-0.1
The graph uses less than half of the given graph-paper area	-0.1
Points are too small or difficult to see	-0.1
The plotting range is inappropriate, e.g. data occupy less than half of the scale	-0.1
Incorrect plotting	subtract as appropriate

Part A : Measurement of the coefficient of restitution (2.3 points)

A.1 (0.2 pt)

Express the coefficient of restitution e in terms of h and h_b .

解答. The speed just before impact is

$$v_0 = \sqrt{2gh},$$

The speed just after rebound is

$$v_1 = \sqrt{2gh_b}$$

therefore

$$e = \frac{v_1}{v_0} = \sqrt{\frac{h_b}{h}}.$$

Thus

$$\eta = \frac{h_b}{h} = e^2$$

.

Marking Scheme

Correctly writes the speeds before and after collision, such as $v_0 = \sqrt{2gh}$ and $v_1 = \sqrt{2gh_b}$	0.1
Obtains $e = \sqrt{h_b/h}$	0.1

A.2 (1.0 pt)

For each drop height, calculate the mean rebound height $\overline{h_b}$ and the uncertainty δh .

解答.

$$\overline{h_b} = \frac{1}{5} \sum_{i=1}^5 h_{b,i}, \quad \delta h = \sqrt{\frac{1}{20} \sum_{i=1}^5 (h_{b,i} - \overline{h_b})^2}.$$

Filling the table on the answer sheet gives, for example, the following. Red values are sample entries.

$$h = 120 \text{ mm}$$

h_b	82	94	104	89	101	94.0
$h_b - \overline{h_b}$	-12.0	0.0	10.0	-5.0	7.0	
$(h_b - \overline{h_b})^2$	144	0	100	25	49	

$h = 180 \text{ mm}$

h_b	129	151	138	148	143	141.8
$h_b - \overline{h_b}$	-12.8	9.2	-3.8	6.2	1.2	
$(h_b - \overline{h_b})^2$	163.84	84.64	14.44	38.44	1.44	

$h = 240 \text{ mm}$

h_b	178	195	208	186	199	193.2
$h_b - \overline{h_b}$	-15.2	1.8	14.8	-7.2	5.8	
$(h_b - \overline{h_b})^2$	231.04	3.24	219.04	51.84	33.64	

$h = 300 \text{ mm}$

h_b	223	241	257	232	247	240.0
$h_b - \overline{h_b}$	-17.0	1.0	17.0	-8.0	7.0	
$(h_b - \overline{h_b})^2$	289	1	289	64	49	

$h = 360 \text{ mm}$

h_b	270	292	304	280	297	288.6
$h_b - \overline{h_b}$	-18.6	3.4	15.4	-8.6	8.4	
$(h_b - \overline{h_b})^2$	345.96	11.56	237.16	73.96	70.56	

Thus

$h \text{ [mm]}$	$\overline{h_b} \text{ [mm]}$	$\delta h \text{ [mm]}$
120	94.0	4.0
180	141.8	3.9
240	193.2	5.2
300	240.0	5.9
360	288.6	6.1

is obtained.

Marking Scheme

Correctly uses the formulas for the mean and deviation 0.2

Correctly obtains $(\overline{h_b}, \delta h)$ for each of $h = 120, 180, 240, 300, 360 \text{ mm}$; 0.16 per pair, up to 5 pairs 0.8

A.3 (0.8 pt)

Determine the energy retention ratio η of the ball before and after the collision by plotting the results of **A.2**. Include error bars and give η with its uncertainty. Assume no uncertainty in h .

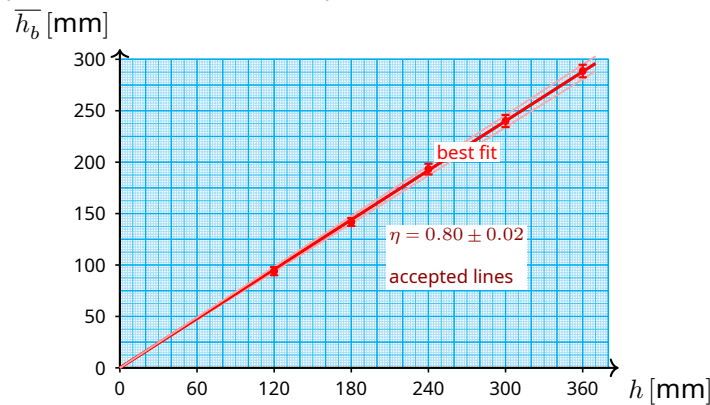
解答.

$$h_b = \eta h$$

Therefore, plotting h on the horizontal axis and $\overline{h_b}$ on the vertical axis and fitting a straight line through the origin gives η as the slope.

In physics olympiad style, the slope is usually read from the graph. In the graph below, the ex-

perimental points and error bars are shown, the best-fit line is drawn as a solid red line, and representative acceptable lines are drawn as pale red lines.



From the graph,

$$\eta \simeq 0.80$$

To estimate the uncertainty, fit through the origin after placing each point at $\bar{h}_b - \delta h$ and then at $\bar{h}_b + \delta h$. These fits give representative lower and upper slopes. The two pale lines in the graph correspond to them.

$$\eta_{\text{low}} \approx 0.780, \quad \eta_{\text{high}} \approx 0.819$$

Therefore

$$\Delta\eta \approx \frac{\eta_{\text{high}} - \eta_{\text{low}}}{2} \approx 0.020$$

and

$$\eta = 0.80 \pm 0.02$$

is a reasonable estimate.

Marking Scheme

Axis labels and ticks, plotted points, error bars, and a fitted line are all present	0.10×4
The central value of η lies in [0.79, 0.81]	0.25
or [0.77, 0.83]	0.15
or [0.75, 0.85]	0.05
$\Delta\eta$ lies in [0.010, 0.030]	0.15
or [0.005, 0.040]	0.10

A.4 (0.3 pt)

Using the result of **A.3**, evaluate the coefficient of restitution e with uncertainty.

解答.

$$e = \sqrt{\eta} = \sqrt{0.80} = 0.894.$$

Using error propagation,

$$\Delta e = \frac{\Delta \eta}{2\sqrt{\eta}}$$

gives

$$\Delta e = \frac{0.02}{2\sqrt{0.80}} \simeq 0.011.$$

Therefore

$$e = 0.894 \pm 0.011$$

is obtained. Rounding to

$$e = 0.89 \pm 0.02$$

is also acceptable.

Marking Scheme

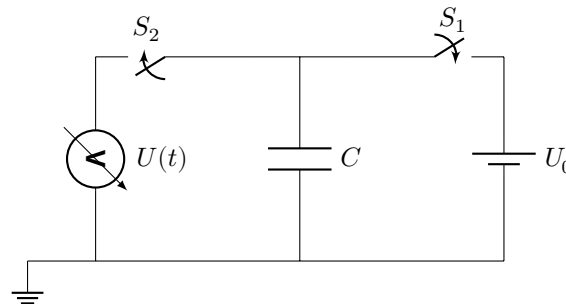
Uses $e = \sqrt{\eta}$	0.05
Uses the error propagation formula $\Delta e = \Delta \eta / (2\sqrt{\eta})$	0.15
Gives e and Δe consistent with η and $\Delta \eta$ obtained in A.3	0.1

Part B : Measurement of the capacitance C (2.9 points)

B.1 (0.7 pt)

Draw a circuit diagram that can perform the charging and discharging process needed to measure the unknown capacitor C , and explain the procedure.

解答. First charge the capacitor with the battery. Then disconnect the battery and connect the voltmeter; the capacitor discharges through the internal resistance R_V of the voltmeter. One possible circuit is shown below.



One possible procedure is as follows.

- Close S_1 and open S_2 to charge the capacitor sufficiently with the battery.
- After charging, open S_1 to disconnect the battery.
- Immediately close S_2 , connect the voltmeter, and start the discharge.
- Read the voltage $U(t)$ at times $t = 0, 5, 10, \dots$

Marking Scheme

The circuit diagram shows an arrangement in which the unknown capacitor, battery, and voltmeter internal resistance are used for discharge 0.3

The procedure clearly describes charging, disconnection, start of discharge, and voltage reading 0.4

B.2 (0.6 pt)

Express the voltage $U(t)$ read by the multimeter using U_0 and any necessary parameters. Show the derivation.

解答. During discharge, a current

$$I = \frac{U}{R_V}$$

flows through the internal resistance R_V of the voltmeter. On the other hand, for the capacitor,

$$I = -C \frac{dU}{dt}$$

hence

$$C \frac{dU}{dt} = -\frac{U}{R_V}.$$

Solving this differential equation gives

$$\frac{dU}{U} = -\frac{dt}{R_V C}$$

and therefore

$$\ln \frac{U(t)}{U_0} = -\frac{t}{R_V C}.$$

Thus

$$U(t) = U_0 \exp\left(-\frac{t}{R_V C}\right)$$

is obtained.

Marking Scheme

Sets up the discharge equation $C dU/dt = -U/R_V$ or an equivalent equation

0.3

Exponential decay: $U(t) = U_0 e^{-t/(R_V C)}$ or derives $\ln(U/U_0) = -t/(R_V C)$

0.3

B.3 (1.6 pt)

Linearize the data in **Table B-1** appropriately, draw a graph, and determine the capacitance C .
Uncertainty analysis is not required.

解答.

$$\ln \frac{U(t)}{U_0} = -\frac{t}{R_V C}$$

it is useful to plot t on the horizontal axis and

$$y = \ln \frac{U}{U_0}$$

on the vertical axis. This gives a straight line. Alternatively,

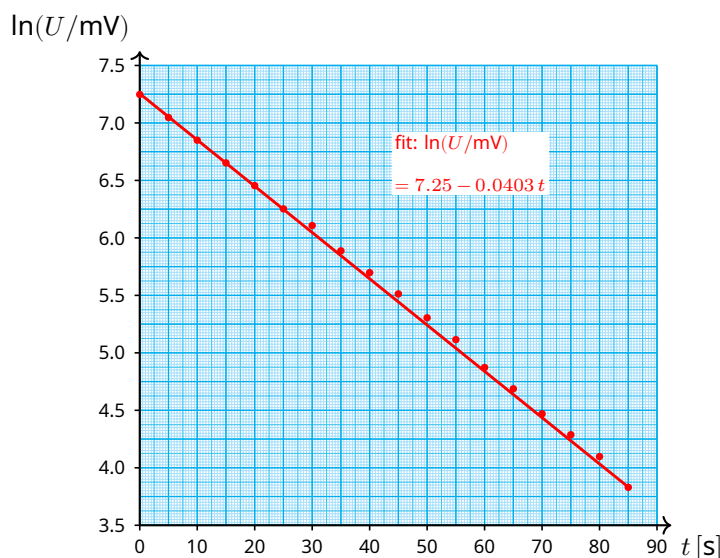
$$y' = \ln \frac{U}{\text{mV}}$$

may be used; this only differs from $\ln(U/U_0)$ by an additive constant, so the slope is unchanged.
Thus an appropriate linearization is to use either $\ln(U/U_0)$ or $\ln(U/\text{mV})$.

One possible transformed table is as follows.

t [s]	0	5	10	15	20	25	30	35	40
U [mV]	1405	1151	943	773	634	518	426	342	283
$\ln(U/U_0)$	0.000	-0.199	-0.399	-0.598	-0.796	-0.998	-1.193	-1.413	-1.602
$\ln(U/\text{mV})$	7.248	7.048	6.849	6.650	6.452	6.250	6.054	5.835	5.645
t [s]	45	50	55	60	65	70	75	80	85
U [mV]	235	191	158	124	103	83	69	57	46
$\ln(U/U_0)$	-1.788	-1.996	-2.185	-2.428	-2.613	-2.829	-3.014	-3.205	-3.419
$\ln(U/\text{mV})$	5.460	5.252	5.063	4.820	4.635	4.419	4.234	4.043	3.829

As an alternative, if $\ln(U/\text{mV})$ is plotted on the vertical axis, the graph looks as follows.



Fitting these data to the straight line

$$\ln \frac{U}{U_0} = at + b$$

gives

$$a \approx -0.0403 \text{ s}^{-1}, \quad b \approx 0$$

Therefore, when using $\ln(U/\text{mV})$, the fit can be written as

$$\ln \frac{U}{\text{mV}} \approx 7.25 - 0.0403 t$$

Hence

$$a = -\frac{1}{R_V C}$$

and therefore

$$C = -\frac{1}{a R_V} = -\frac{1}{(-0.0403) \times 11.1 \times 10^6} \approx 2.24 \times 10^{-6} \text{ F.}$$

Thus

$$C \approx 2.24 \mu\text{F}$$

.

Marking Scheme

Shows a linearization using $\ln(U/U_0)$ or $\ln(U/\text{mV})$	0.4
Correctly fills the transformed table; subtract 0.05 for each transformation error	0.5
Graph has axes t and a logarithmic quantity, axis labels, ticks, at least 9 points, and a fitted line	0.3
The value of C lies in $[2.20, 2.28] \mu\text{F}$	0.4
or $[2.16, 2.32] \mu\text{F}$	0.25
or $[2.00, 2.48] \mu\text{F}$	0.15

Part C : Measurement of the capacitance C_p of the piezoelectric element (2.9 points)

C.1 (0.6 pt)

Using **Figure 2**, write the complete measurement sequence for obtaining V_n on the answer sheet. Distinguish the initial discharge, the charge-transfer loop repeated n times, and the final voltage measurement. Write each step in the form "set/keep S_i open/closed", and make clear that the known capacitor C is not discharged during the loop.

解答. The complete measurement sequence can be written, for example, as follows.

1. At the start of the measurement, open S_1, S_2, S_4 and keep S_3 closed long enough to discharge the known capacitor C , so that $V_0 = 0$. Then open S_3 as well.
2. Repeat the following operation n times.

Step in loop	Operation
1	Keep S_2, S_3, S_4 open and close S_1 to charge C_p to U_0
2	Open S_1 to disconnect the piezoelectric element from the battery
3	Close S_2 to connect C_p and C , and transfer charge
4	Open S_2 again. At this moment a voltage remains on C
5	Close S_4 to discharge only C_p , and then open S_4 again

3. After completing the above loop n times, close S_3 briefly and read the voltage V_n of the known capacitor.

During this measurement, S_3 is closed only during the initial discharge and the final voltage measurement. During the intermediate n loops, keep S_3 open and do not discharge the known capacitor C .

Marking Scheme

Discharges the known capacitor C at the beginning and sets $V_0 = 0$	0.1
Uses the correct order: charge piezoelectric element, disconnect battery, transfer charge to C , discharge C_p	0.3
Keeps S_3 open during the loop so that the known capacitor C is not discharged	0.1
States that V_n is read only after repeating the loop n times	0.1

C.2 (0.6 pt)

Let V_n be the voltage appearing across the known capacitor C after n loops. Express V_n in terms of the battery voltage U_0 , C , C_p , and n . Show the derivation. Assume that $V_0 = 0$.

解答. After $(n - 1)$ loops, the known capacitor C has charge

$$Q_C = CV_{n-1}$$

. On the other hand, in each loop the piezoelectric element is charged to

$$Q_p = C_p U_0$$

before being disconnected from the battery.

When S_2 is then closed, C and C_p are connected in parallel and reach a common voltage V_n .

Charge conservation gives

$$C V_{n-1} + C_p U_0 = (C + C_p) V_n.$$

Therefore

$$V_n = \frac{C}{C + C_p} V_{n-1} + \frac{C_p}{C + C_p} U_0$$

is obtained.

Now set

$$W_n = U_0 - V_n$$

. Then

$$W_n = \frac{C}{C + C_p} W_{n-1}$$

and since $V_0 = 0$, $W_0 = U_0$. Hence

$$W_n = U_0 \left(\frac{C}{C + C_p} \right)^n.$$

Therefore

$$V_n = U_0 \left[1 - \left(\frac{C}{C + C_p} \right)^n \right]$$

is obtained.

Marking Scheme

Recursion relation $V_n = \frac{C}{C + C_p} V_{n-1} + \frac{C_p}{C + C_p} U_0$ or an equivalent equation 0.3

Exact formula $V_n = U_0 \left[1 - \left(\frac{C}{C + C_p} \right)^n \right]$ is derived 0.3

C.3 (1.7 pt)

Using **Table C-1**, draw an appropriate graph and determine the capacitance C_p of the piezoelectric element. Use the values of U_0 and C obtained in Part B. You may use $C \gg C_p$ if needed.

解答. Use the Part B results

$$U_0 = 1405 \text{ mV}, \quad C = 2.24 \mu\text{F}$$

.

Exact method. From the result of **C.2**,

$$U_0 - V_n = U_0 \left(\frac{C}{C + C_p} \right)^n$$

. Therefore

$$\ln(U_0 - V_n) = \ln U_0 + n \ln \left(\frac{C}{C + C_p} \right)$$

. Hence, plotting n on the horizontal axis and

$$y = \ln(U_0 - V_n)$$

or

$$y' = \ln \left(\frac{U_0 - V_n}{\text{mV}} \right)$$

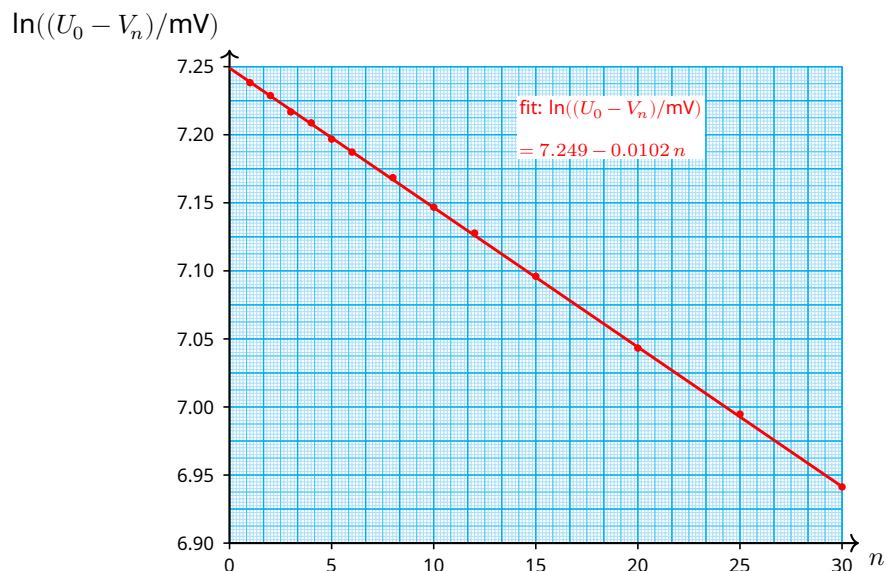
on the vertical axis gives a straight line.

Filling the table on the answer sheet gives, for example, the following.

n	1	2	3	4	5	6	8
V_n [mV]	13.3	26.5	43.0	53.9	69.8	82.5	107
$\ln((U_0 - V_n)/\text{mV})$	7.238	7.229	7.217	7.209	7.197	7.187	7.169

n	10	12	15	20	25	30
V_n [mV]	135	159	198	260	314	371
$\ln((U_0 - V_n)/\text{mV})$	7.147	7.128	7.096	7.043	6.995	6.941

The corresponding graph is shown below.



Fitting the straight line

$$\ln \left(\frac{U_0 - V_n}{\text{mV}} \right) = an + b$$

gives

$$a \approx -0.0102, \quad b \approx 7.249$$

Here

$$a = \ln \left(\frac{C}{C + C_p} \right)$$

. Thus

$$\frac{C}{C + C_p} = e^a.$$

Therefore

$$C_p = C(e^{-a} - 1)$$

. Substituting the numerical values gives

$$C_p = 2.24 \times 10^{-6} (e^{0.0102} - 1) \approx 2.30 \times 10^{-8} \text{ F.}$$

Hence

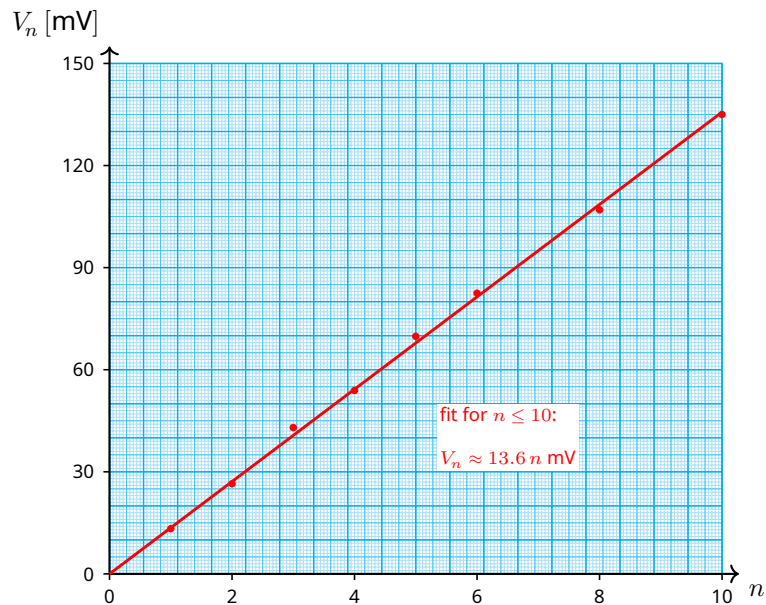
$$C_p \approx 23.0 \text{ nF}$$

.

Approximate method. In the initial region where $C \gg C_p$ and $nC_p \ll C$,

$$V_n \approx n \frac{C_p}{C} U_0$$

so one may plot V_n against n and determine the slope using only the first few points. However, large n data clearly deviate from a straight line, so it is inappropriate to fit all data with this approximation. If only the data with $n \leq 10$ in **Table C-1** are used, the graph below is approximately linear.



Let the slope of this line be

$$s = \frac{dV_n}{dn}$$

. Then

$$s \approx \frac{C_p}{C} U_0$$

and therefore

$$C_p \approx \frac{sC}{U_0}$$

. Reading from the graph,

$$s \approx 13.6 \text{ mV}$$

gives

$$C_p \approx \frac{13.6}{1405} \times 2.24 \mu\text{F} \approx 2.16 \times 10^{-8} \text{ F.}$$

Thus the approximate method gives

$$C_p \approx 21.6 \text{ nF} \approx 22 \text{ nF}$$

which is also reasonable.

Marking Scheme

Shows a linearization using $\ln(U_0 - V_n)$ or $\ln((U_0 - V_n)/\text{mV})$, or uses $V_n \propto n$ in the initial region	0.6
For the exact method, correctly fills the transformed table; for the approximate method, states that the fit is restricted to about $n \leq 10$ and that all points should not be fitted	0.3
Graph has axis labels, ticks, at least 8 points, and a fitted line	0.4
C_p lies in $[21, 24] \text{ nF}$	0.4
or $[20, 25] \text{ nF}$	0.25
or $[18, 27] \text{ nF}$	0.15

--

Part D : Force-voltage response of the piezoelectric element (5.0 points)

D.1 (0.4 pt)

Give 2 reasons why directly connecting the piezoelectric element to a voltmeter and reading the voltage does not work well. One of the reasons must use the term time constant.

解答. The first reason is that the piezoelectric capacitance C_p and the internal resistance R_V of the voltmeter form a discharge time constant

$$\tau = R_V C_p$$

. For example,

$$\tau \approx (11.1 \times 10^6)(23 \times 10^{-9}) \approx 0.26 \text{ s}$$

, so the value decreases significantly before it can be read.

The second reason is that it is difficult to isolate the force corresponding to one particular instant of the scale reading. While the force is being applied, the voltage changes and discharge proceeds, so a direct connection cannot stably associate a single instantaneous force with V .

Marking Scheme

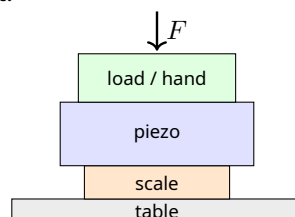
Explains using the time constant $\tau = R_V C_p$ that a direct connection causes discharge 0.3

States another valid reason, such as difficulty matching an instantaneous scale reading to the piezoelectric voltage or the fact that the value changes while the force is applied 0.1

D.2 (1.1 pt)

Draw a simple diagram showing the arrangement of the piezoelectric element, the scale, and the way the force is applied. Then list the measurement procedure using the circuit in **Figure 3**. The procedure must make it possible to read the value of V corresponding to a particular instantaneous value of the force F applied to the piezoelectric element, and must include the operation for discharging the piezoelectric element and the capacitor before measuring another value. Briefly explain why U must be read quickly after opening/closing the switch for voltage readout.

解答. One possible arrangement is shown below. By stacking the scale and the piezoelectric element vertically in series, the force indicated by the scale can be treated as the same force applied to the piezoelectric element.



One possible measurement procedure is as follows.

1. Initially open S_1, S_3 and close S_2, S_4 to discharge the piezoelectric element and the known capacitor.
2. Just before measurement, open S_2, S_4 again.
3. With the scale and piezoelectric element stacked in series, apply a load from above and read the scale value M when it is stable.
4. While keeping that force, close S_1 briefly and transfer the charge of the piezoelectric element to the known capacitor C .
5. Immediately open S_1 , then close S_3 and read the voltage U of the known capacitor.
6. After the reading, open S_3 and close S_2, S_4 to discharge the piezoelectric element and the known capacitor again.
7. Change the load and repeat the same procedure.

U must be read quickly because, when S_3 is closed, the known capacitor C begins to discharge through the internal resistance of the voltmeter, so U decreases immediately.

Marking Scheme

Draws an arrangement in which the scale and piezoelectric element are in series and carry the same force	0.2
Correctly describes initial discharge, charge transfer while force is applied, readout, and discharge before the next measurement	0.6
Explains that U decreases if not read quickly because the capacitor discharges	0.3

D.3 (0.3 pt)

In the circuit of **Figure 3**, give the formula relating the readout voltage U to the voltage V of the piezoelectric element. Also give the formula relating the scale reading M to the force F .

解答. The charge on the piezoelectric element is

$$Q = C_p V$$

. After this charge is transferred to the initially uncharged known capacitor C ,

$$Q = (C + C_p)U$$

. Therefore

$$C_p V = (C + C_p)U$$

that is,

$$V = \frac{C + C_p}{C_p} U \approx \frac{C}{C_p} U$$

.

If the scale reading is M , then

$$F = Mg$$

. Here M must be used in kg. If the display is in g, convert it to kg by using $M/1000$.

Marking Scheme

Correctly writes $V = \frac{C + C_p}{C_p} U$ or the approximation $V \approx \frac{C}{C_p} U$ 0.2

Correctly writes $F = Mg$ 0.1

D.4 (1.8 pt)

Convert the data in **Table D-1** to F and V , and plot them on a graph covering a sufficiently wide range. You do not need to convert and plot all data points. Choose points so that the overall trend is clear.

解答. Using the result of **D.3**,

$$F = Mg, \quad V = \frac{C + C_p}{C_p} U$$

. From Parts B and C,

$$C = 2.24 \mu\text{F}, \quad C_p \approx 23.0 \text{ nF}$$

which gives

$$\frac{C + C_p}{C_p} \approx 98.4$$

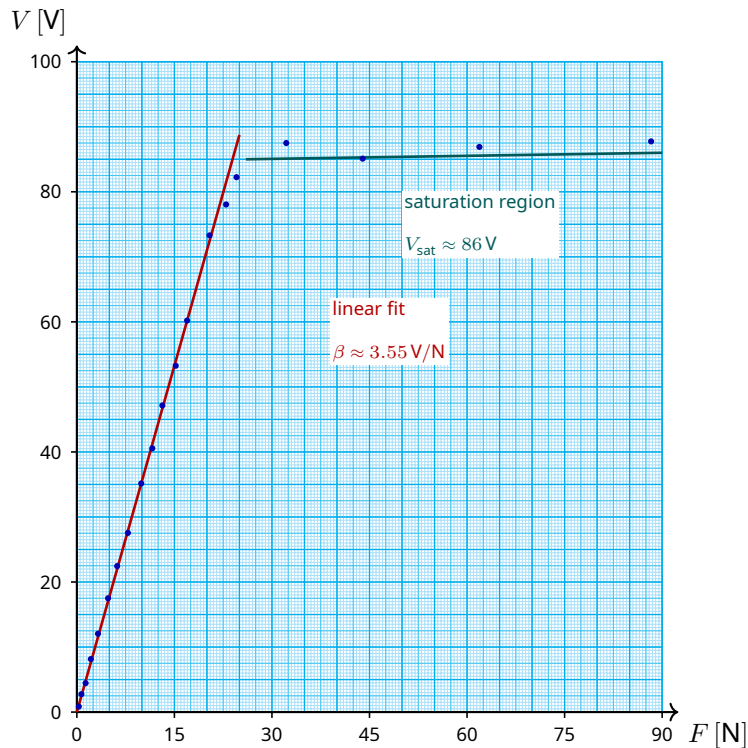
.

To include many points in the low-force linear region while also not ignoring the saturation region, one may choose and convert, for example, the following 20 points.

M [kg]	U [mV]	F [N]	V [V]
0.030	8.6	0.29	0.8
0.070	27.8	0.69	2.7
0.140	45.3	1.37	4.5
0.220	82.7	2.16	8.1
0.330	122.6	3.24	12.1
0.490	178.0	4.81	17.5
0.630	228	6.18	22.4
0.800	280	7.85	27.6
1.01	357	9.91	35.1
1.18	412	11.58	40.5

M [kg]	U [mV]	F [N]	V [V]
1.34	479	13.15	47.1
1.55	541	15.21	53.2
1.73	612	16.97	60.2
2.08	745	20.40	73.3
2.34	793	22.96	78.0
2.50	836	24.53	82.3
3.28	889	32.18	87.5
4.48	865	43.95	85.1
6.31	883	61.90	86.9
9.00	892	88.29	87.8

Plotting these points shows that the response is approximately linear at small forces and that V saturates to an almost constant value at large forces.



Marking Scheme

At least 8 selected points are from the linear region $F \leq 15$ N	0.2
At least 1 point is selected from $F \geq 60$ N, and at least 3 points with $F \geq 20$ N are included	0.2
At least 16 data points are converted and written	0.6
or at least 12 data points	0.3
Graph has axis labels, units, ticks, a linear scale, and at least 10 points	0.4
Selected points are plotted approximately correctly in the F - V plane	0.2

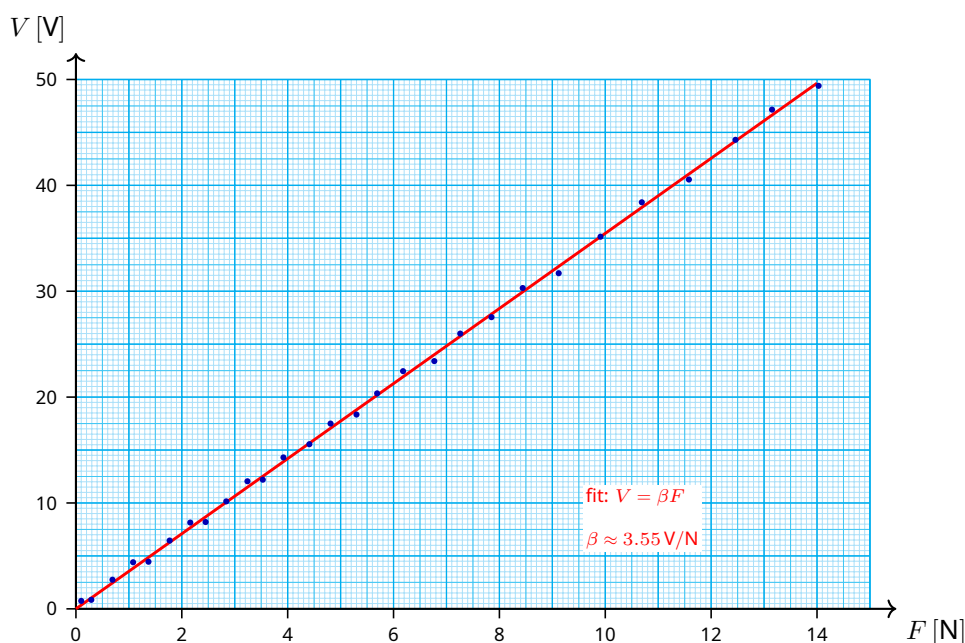
D.5 (0.8 pt)

For small forces, $V = \beta F$ holds. Choose the linear region appropriately, plot it again on a separate graph, and determine the proportionality constant β .

解答. Choose a small-force range, for example

$$M \leq 1.43 \text{ kg} \quad (F \lesssim 15 \text{ N})$$

as the linear region. Enlarging only the points in this range gives the graph below.



Fitting with a straight line through the origin,

$$V = \beta F$$

gives

$$\beta \approx 3.55 \text{ V/N}$$

Therefore

$$\beta \approx 3.55 \text{ V/N}$$

.

Marking Scheme

At least 8 points with, for example, $F \leq 20 \text{ N}$ are selected and plotted separately as the linear region 0.4

The value of β lies in $[3.40, 3.75] \text{ V/N}$ 0.4

or $[3.20, 3.95] \text{ V/N}$ 0.25

or $[3.00, 4.20] \text{ V/N}$ 0.15

D.6 (0.6 pt)

Using the results of **D.4** and **D.5**, together with C_p obtained in Part C, calculate the saturation voltage V_{sat} , the corresponding force F_{sat} , the saturation pressure

$$p_{\text{sat}} = \frac{F_{\text{sat}}}{A}$$

and the maximum surface charge density

$$\sigma_{\max} = \frac{C_p V_{\text{sat}}}{A}$$

.

解答. From the graph in **D.4**, in the saturation region

$$V_{\text{sat}} \approx 86 \text{ V}$$

can be read.

Using the result $\beta \approx 3.55 \text{ V/N}$ from **D.5**,

$$F_{\text{sat}} = \frac{V_{\text{sat}}}{\beta} \approx \frac{86}{3.55} \approx 24.2 \text{ N}.$$

Therefore

$$p_{\text{sat}} = \frac{F_{\text{sat}}}{A} \approx \frac{24.2}{1.33 \times 10^{-4}} \approx 1.82 \times 10^5 \text{ Pa}.$$

Also, using $C_p \approx 23.0 \text{ nF}$,

$$\sigma_{\max} = \frac{C_p V_{\text{sat}}}{A} \approx \frac{(23.0 \times 10^{-9}) \times 86}{1.33 \times 10^{-4}} \approx 1.49 \times 10^{-2} \text{ C/m}^2.$$

Hence

$$V_{\text{sat}} \approx 86 \text{ V}, \quad F_{\text{sat}} \approx 24.2 \text{ N},$$

$$p_{\text{sat}} \approx 1.8 \times 10^5 \text{ Pa}, \quad \sigma_{\max} \approx 1.5 \times 10^{-2} \text{ C/m}^2.$$

Marking Scheme

V_{sat} lies in [67, 101] V	0.15
F_{sat} lies in [18.8, 28.2] N	0.15
p_{sat} lies in $[1.4, 2.1] \times 10^5 \text{ Pa}$	0.15
$\sigma_{\max}$ lies in $[1.2, 1.8] \times 10^{-2} \text{ C/m}^2$	0.15

Part E : Ball deformation and contact stress (4.5 points)

E.1 (0.3 pt)

Let $x_{\max}$ be the deformation when the ball is maximally compressed during the collision. Express the maximum voltage $V_{\max}$ indicated by the piezoelectric element in terms of $x_{\max}$.

解答. At maximum deformation,

$$F_{\max} = kx_{\max}^\alpha$$

and using the linear calibration from Part D,

$$V = \beta F$$

gives

$$V_{\max} = \beta F_{\max} = \beta k x_{\max}^{\alpha}.$$

Therefore

$$V_{\max} = \beta k x_{\max}^{\alpha}$$

.

Marking Scheme

Uses $F_{\max} = kx_{\max}^{\alpha}$

0.15

Correctly combines it with $V_{\max} = \beta F_{\max}$

0.15

E.2 (0.3 pt)

Consider the process in which the ball rebounds after deforming up to $x_{\max}$. By considering the work

$$\int F dx$$

done by the normal force on the ball, express the elastic energy E_{el} stored up to the instant of rebound in terms of k , $x_{\max}$, and α .

解答.

$$E_{\text{el}} = \int_0^{x_{\max}} F dx = \int_0^{x_{\max}} kx^{\alpha} dx = \frac{k}{\alpha + 1} x_{\max}^{\alpha+1}.$$

Hence

$$E_{\text{el}} = \frac{k}{\alpha + 1} x_{\max}^{\alpha+1}.$$

Marking Scheme

Sets up the integral correctly

0.15

Obtains $\frac{k}{\alpha + 1} x_{\max}^{\alpha+1}$

0.15

E.3 (0.5 pt)

Assume that the elastic energy E_{el} equals the initial mechanical energy multiplied by the energy retention ratio η obtained in Part A,

$$E_{\text{el}} = \eta mgh$$

. If the relation between the maximum voltage $V_{\max}$ and the drop height h is

$$V_{\max} = ch^{\gamma}$$

, express the exponent γ in terms of α .

解答. From E.2,

$$\frac{k}{\alpha + 1} x_{\max}^{\alpha+1} = \eta m g h$$

so

$$x_{\max} \propto h^{1/(\alpha+1)}.$$

Substituting this into the result of E.1,

$$V_{\max} = \beta k x_{\max}^{\alpha}$$

gives

$$V_{\max} \propto h^{\alpha/(\alpha+1)}.$$

Therefore

$$\gamma = \frac{\alpha}{\alpha + 1}.$$

Marking Scheme

Reaches $x_{\max} \propto h^{1/(\alpha+1)}$

0.2

Uses $V_{\max} \propto h^{\alpha/(\alpha+1)}$

0.15

Obtains $\gamma = \frac{\alpha}{\alpha + 1}$

0.15

E.4 (0.5 pt)

Express the relation between the proportionality constant c and k using the necessary parameters.

解答. From the equation in E.2,

$$x_{\max} = \left(\frac{(\alpha + 1) \eta m g h}{k} \right)^{1/(\alpha+1)}$$

. Substituting this into E.1 gives

$$V_{\max} = \beta k \left(\frac{(\alpha + 1) \eta m g h}{k} \right)^{\alpha/(\alpha+1)}.$$

Therefore

$$V_{\max} = c h^{\gamma}$$

has coefficient

$$c = \beta k^{1/(\alpha+1)} ((\alpha + 1) \eta m g)^{\alpha/(\alpha+1)}.$$

Marking Scheme

Expresses $x_{\max}$ in terms of h

0.2

Arranges the result in the form $V_{\max} = c h^{\gamma}$

0.15

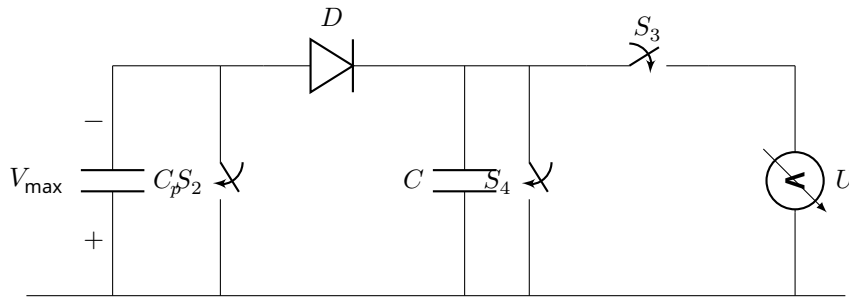
Correct relation between c and k

0.15

E.5 (0.5 pt)

To measure $V_{\max}$ during the collision process, where and in which direction should a diode be inserted in the circuit of **Figure 3**? Draw the complete circuit diagram.

解答. The diode should be inserted in series in the branch from the piezoelectric element C_p to the known capacitor C . Its orientation should allow charge to flow from the piezoelectric element to the capacitor during collision and prevent reverse flow.



In other words, insert it with **the piezoelectric side as the anode and the capacitor side as the cathode**. This holds the charge corresponding to the maximum voltage during the collision on C .

Marking Scheme

Places the diode in the branch between C_p and C	0.3
Diode direction allows current from the piezoelectric element to the capacitor	0.2

E.6 (1.6 pt)

Determine the value of α from the data in **Table E-1**. Use η from Part A, β from Part D, and the results of Parts B and C if needed.

解答. For each height, calculate the average

$$\bar{U} = \frac{1}{5} \sum_{i=1}^5 U_i$$

. Also, from Parts B and C,

$$\frac{C + C_p}{C_p} \approx 98.4$$

so

$$V_{\max} = \frac{C + C_p}{C_p} \bar{U}$$

. If $\bar{U}$ is substituted in mV, this may be written as

$$V_{\max}[\text{V}] \approx 0.0984 \bar{U}[\text{mV}]$$

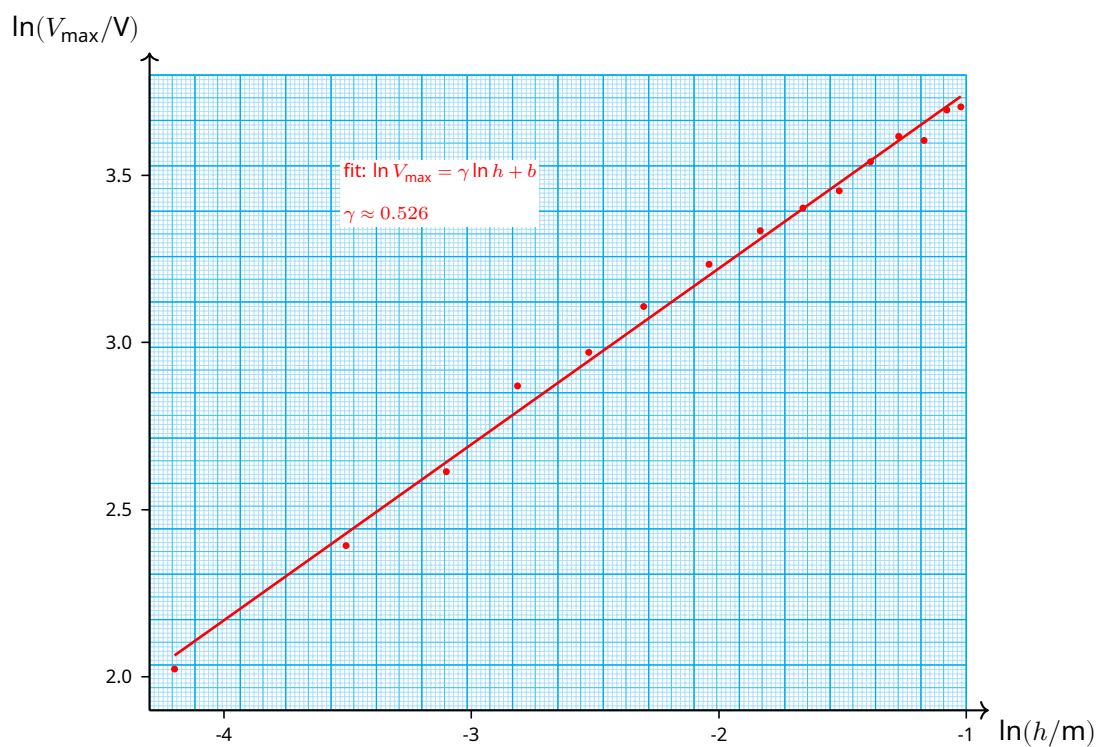
.

Here express h in m and use

$$\ln V_{\max} = \ln c + \gamma \ln h$$

to linearize the data. If $\ln(h/\text{mm})$ is used instead, the slope γ is unchanged.

h [mm]	$\bar{U}$ [mV]	$V_{\max}$ [V]	$\ln(h/\text{m})$	$\ln(V_{\max}/\text{V})$
15	76.8	7.56	−4.200	2.023
30	111.2	10.94	−3.507	2.393
45	138.8	13.66	−3.101	2.614
60	179.4	17.65	−2.813	2.871
80	198.4	19.52	−2.526	2.972
100	227.2	22.36	−2.303	3.107
130	257.8	25.37	−2.040	3.234
160	285.4	28.08	−1.833	3.335
190	305.2	30.03	−1.661	3.402
220	321.2	31.61	−1.514	3.453
250	350.4	34.48	−1.386	3.540
280	378.2	37.21	−1.273	3.617
310	373.8	36.78	−1.171	3.605
340	409.0	40.25	−1.079	3.695
360	413.2	40.66	−1.022	3.705



Fitting this graph with a straight line gives, for example,

$$\ln V_{\max} \approx 0.526 \ln h + 4.27 \quad (h \text{ in m})$$

. Hence

$$\gamma \approx 0.526$$

. Using the result of **E.3**,

$$\alpha = \frac{\gamma}{1 - \gamma}$$

gives

$$\alpha \approx \frac{0.526}{1 - 0.526} \approx 1.11.$$

Therefore

$$\alpha \approx 1.11$$

.

Marking Scheme

Shows an appropriate linearization, such as using $\ln V_{\max}$ and $\ln h$, or using $V_{\max}^{1/\gamma}$ and h	0.4
Correctly converts to $\bar{U}$ and $V_{\max}$, or equivalent quantities	0.4
Draws a graph with axis labels, ticks, at least 12 points, and a fitted line	0.4
α lies in $[0.95, 1.20]$	0.4
or $[0.85, 1.30]$	0.25
or $[0.70, 1.45]$	0.15

E.7 (0.8 pt)

Determine the value of k . The unit may be written as $\text{N m}^{-\alpha}$. You may draw another graph if needed. Use the necessary constants from the previous parts.

解答. From the intercept of the fit in **E.6**,

$$c \approx e^{4.27} \approx 71.8$$

is obtained. Here

$$V_{\max} = ch^\gamma$$

uses h in m. If h is fitted in mm, the corresponding coefficient c_{mm} must be converted by

$$c = c_{\text{mm}} 10^{3\gamma}$$

to return to SI units.

From the formula in **E.4**,

$$c = \beta k^{1/(\alpha+1)} ((\alpha+1)\eta mg)^{\alpha/(\alpha+1)}$$

we have

$$k = \left[\frac{c}{\beta ((\alpha+1)\eta mg)^{\alpha/(\alpha+1)}} \right]^{\alpha+1}.$$

Substituting

$$\alpha \approx 1.11, \quad \eta \approx 0.80, \quad \beta \approx 3.55 \text{ V/N}, \quad m = 8.0 \text{ g}, \quad g = 9.81 \text{ m s}^{-2}$$

gives

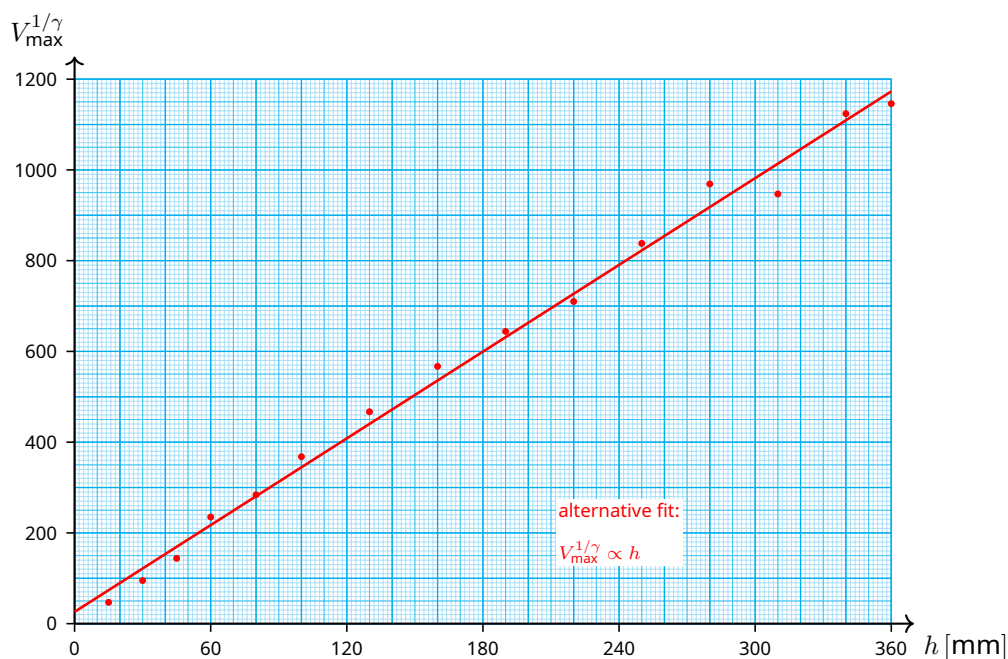
$$k \approx 5.4 \times 10^3 \text{ N m}^{-\alpha}.$$

Therefore

$$k \approx 5.4 \times 10^3 \text{ N m}^{-\alpha}$$

.

As an alternative, plotting $V_{\max}^{1/\gamma}$ against h to determine c is also acceptable.



Marking Scheme

Uses the intercept c , or evaluates c by an equivalent linearization such as $V_{\max}^{1/\gamma}$ vs h	0.3
Correctly uses the relation between c and k	0.2
k lies in $[3.0, 7.0] \times 10^3 \text{ N m}^{-\alpha}$	0.3
or $[2.0, 9.0] \times 10^3 \text{ N m}^{-\alpha}$	0.2
or $[1.0, 12] \times 10^3 \text{ N m}^{-\alpha}$	0.1

Part F : Contact time (2.4 points)

F.1 (0.6 pt)

A ball with coefficient of restitution e is dropped from height h , and the maximum readout voltage is $U_{\max}$. Express τ_{eff} using known parameters and $U_{\max}$. Use the results of Parts C and D if needed.

解答. The speed just before impact is

$$v = \sqrt{2gh}$$

and the speed just after rebound is

$$ev$$

. Therefore the magnitude of the momentum change is

$$\Delta p = m(v + ev) = m(1 + e)\sqrt{2gh}$$

.

On the other hand, from Part D,

$$V_{\max} = \frac{C + C_p}{C_p} U_{\max}, \quad F_{\max} = \frac{V_{\max}}{\beta}$$

hence

$$F_{\max} = \frac{C + C_p}{\beta C_p} U_{\max}.$$

Therefore

$$\tau_{\text{eff}} = \frac{\Delta p}{F_{\max}} = \frac{m(1 + e)\beta\sqrt{2gh}}{((C + C_p)/C_p) U_{\max}}.$$

Here $U_{\max}$ is used in V. If $U_{\max}$ is used in mV, then

$$\tau_{\text{eff}}[\text{S}] = \frac{1000 m(1 + e)\beta\sqrt{2gh}}{((C + C_p)/C_p) U_{\max}[\text{mV}]}$$

.

Marking Scheme

Correctly obtains $\Delta p = m(1 + e)\sqrt{2gh}$

0.3

Uses $F_{\max} = \frac{C + C_p}{\beta C_p} U_{\max}$ and correctly arranges $\tau_{\text{eff}} = \Delta p / F_{\max}$

0.3

F.2 (0.7 pt)

Using the value of e obtained in Part A, calculate τ_{eff} for each height from the data in **Table E-1**. Use the average of the five measured values at each height. Also determine the speed just before collision, $v = \sqrt{2gh}$.

解答. From Part A, use

$$e \approx 0.894$$

. Also

$$\frac{C + C_p}{C_p} \approx 98.4, \quad \beta \approx 3.55 \text{ V/N}$$

.

For each height, average the measured values of U and calculate τ_{eff} from the formula in **F.1**. The result is as follows.

h [mm]	$\bar{U}$ [mV]	v [m/s]	τ_{eff} [ms]
15	76.8	0.542	3.86
30	111.2	0.767	3.77
45	138.8	0.940	3.70
60	179.4	1.085	3.31
80	198.4	1.253	3.45
100	227.2	1.401	3.37
130	257.8	1.597	3.39
160	285.4	1.772	3.39
190	305.2	1.931	3.46
220	321.2	2.078	3.54
250	350.4	2.215	3.46
280	378.2	2.344	3.39
310	373.8	2.466	3.61
340	409.0	2.583	3.45
360	413.2	2.658	3.52

Marking Scheme

Uses $v = \sqrt{2gh}$ and correctly calculates v for at least 10 rows 0.3

Uses the average $\bar{U}$ of the five measurements at each height and correctly calculates τ_{eff} for at least 10 rows 0.4

F.3 (0.5 pt)

For **Table F-1**, do the same calculation as in **F.2**.

解答. For this ball, $e = 0$, so

$$\Delta p = m\sqrt{2gh}$$

. Averaging U at each height and substituting into the formula in **F.1** gives

h [mm]	$\bar{U}$ [mV]	v [m/s]	τ_{eff} [ms]
15	72.6	0.542	2.16
30	113.8	0.767	1.95
70	175.2	1.172	1.93
110	232.0	1.469	1.83
130	271.4	1.597	1.70
170	311.6	1.826	1.69
190	331.6	1.931	1.68
210	348.0	2.030	1.68
250	388.4	2.215	1.65
290	396.6	2.385	1.74
330	452.8	2.545	1.62

Marking Scheme

Uses the formula with $e = 0$

0.1

Correctly calculates v for at least 7 rows

0.15

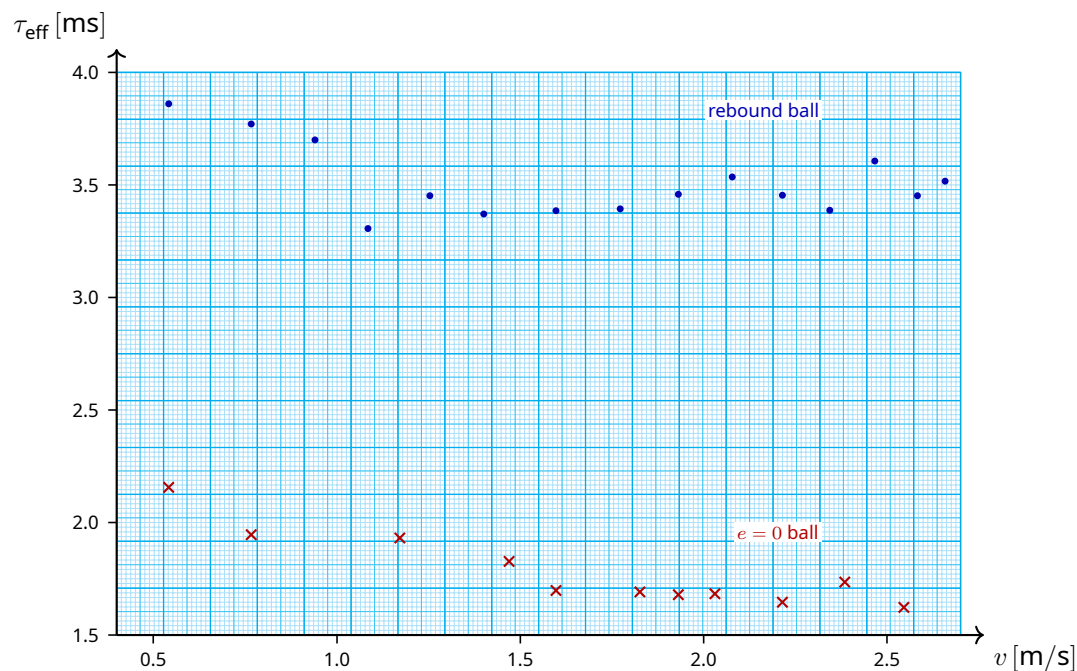
Correctly calculates τ_{eff} for at least 7 rows

0.25

F.4 (0.6 pt)

For the two balls, plot the relation between τ_{eff} and v , and state whether τ_{eff} becomes longer, shorter, or remains almost unchanged as v increases.

解答. Plotting the results of **F.2** and **F.3** on the same graph gives, for example, the following.



For both balls, τ_{eff} does not change greatly with v ; at most it decreases slightly. The ball with larger rebound also has a clearly longer contact time than the ball with $e = 0$.

Marking Scheme

- | | |
|--|-----|
| Plots the two balls on the same graph with distinguishable symbols, axis labels, and at least 7 points for each ball | 0.4 |
| Concludes that the contact time is almost constant or decreases slightly | 0.2 |